CS177 Bioinformatics: Software Development and Applications

Lecture 1: Basics of Molecular Biology

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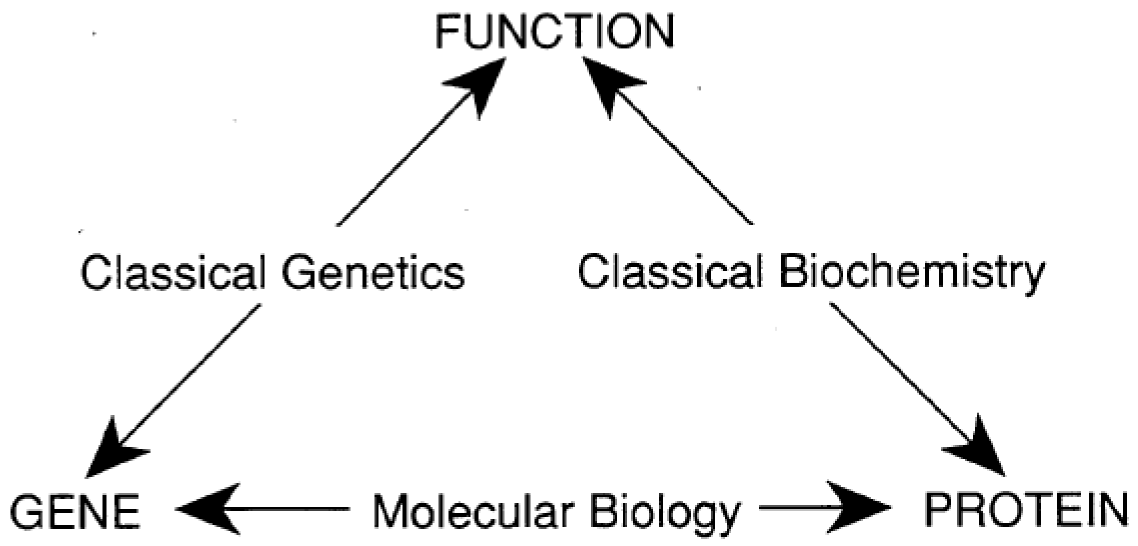
Spring, 2025

Learning objectives

- At the end of this lecture, you will be able to:
  - Explain basic terminology of molecular biology
  - Describe the basic building blocks of life
  - Describe the Central Dogma of Molecular Biology, i.e. how information is transmitted from DNA to proteins
  - Describe the basic processes of RNA transcription and protein translation

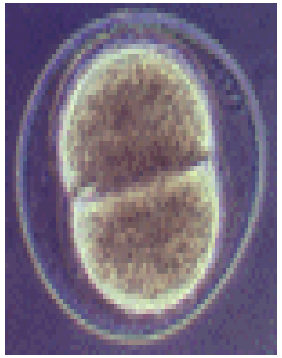
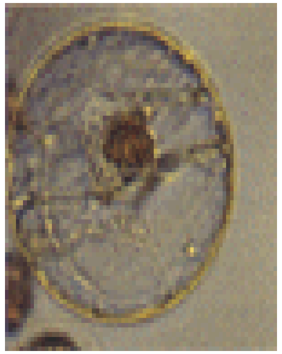
Historical roots of molecular biology

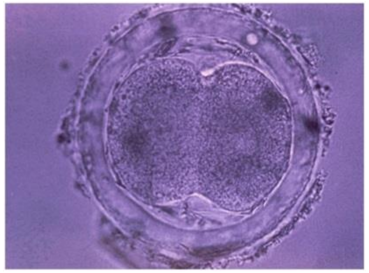
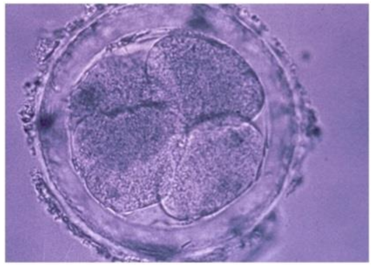
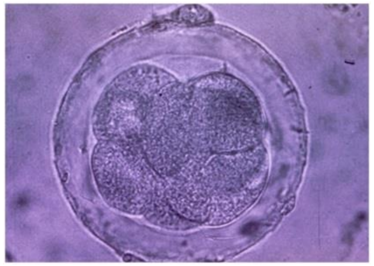


From Fig. 1.10, Lander and Waterman's chapter (1995)

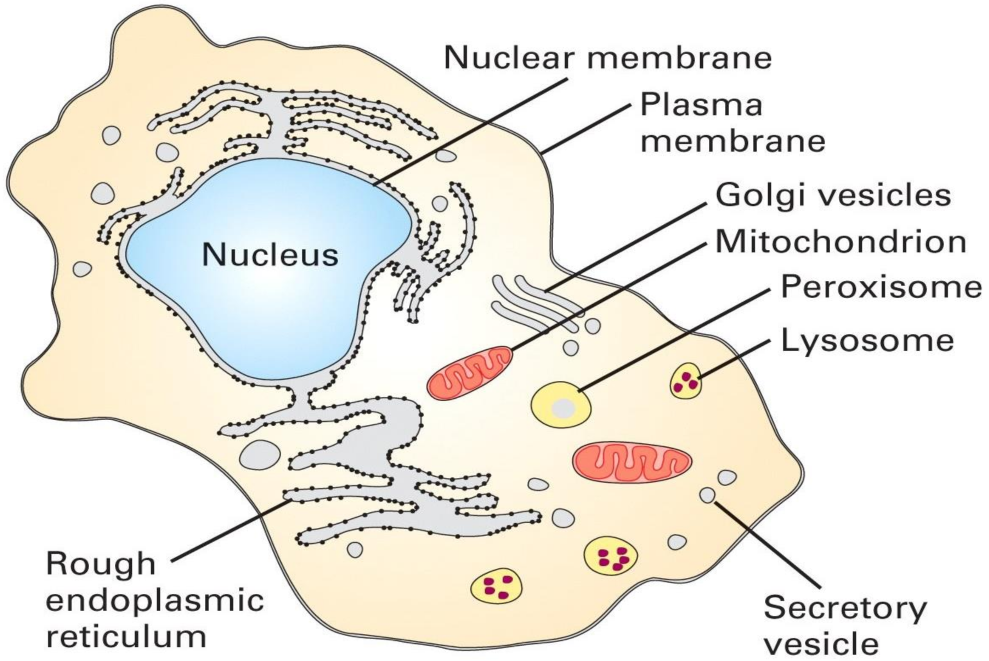
- **Molecular biology** grew out of two experimental approaches to studying biological function: "Biochemistry" and "Genetics"
  - **Biochemistry**: break up the molecules in a living organism, with the goal of purifying and characterizing the chemical components responsible for carrying out a particular function
  - **Genetics**: study mutant organisms that are intact except for a single component

What Is Life Made of?

Life begins with Cell

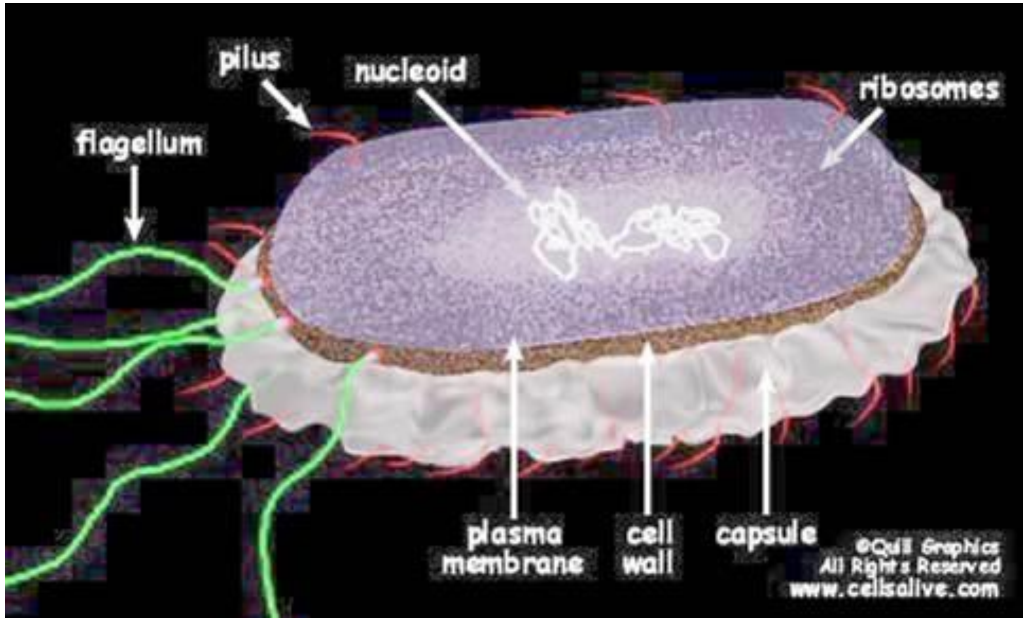


- A cell is a smallest structural unit of an organism that is capable of independent functioning
- All cells have some common features

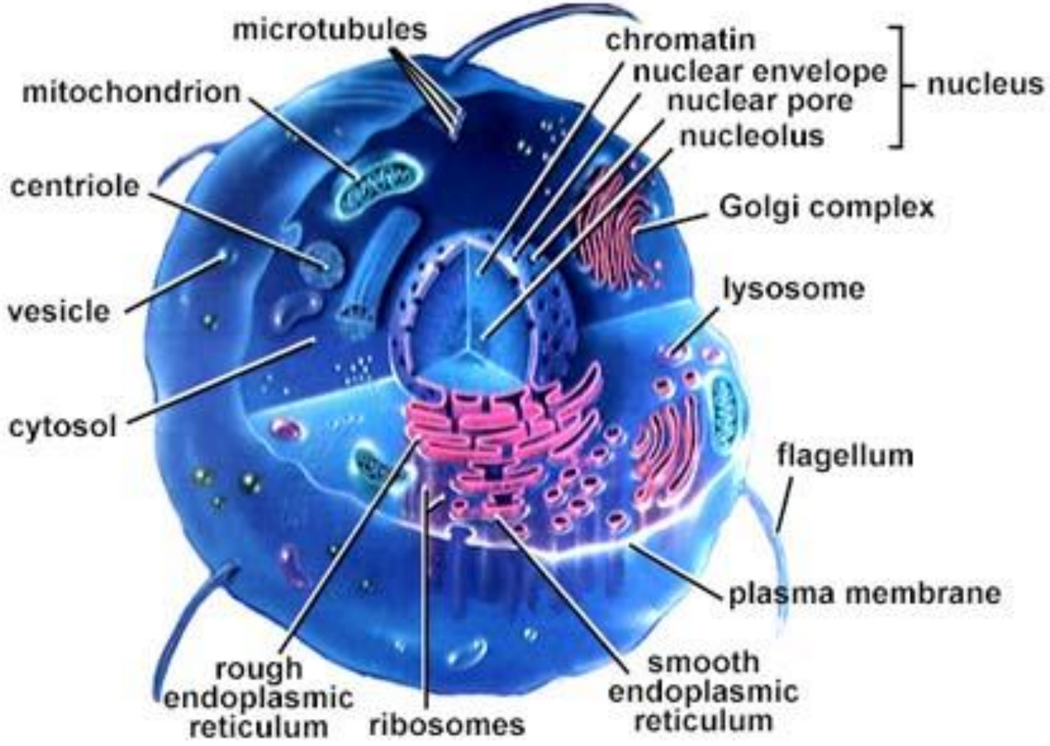
What is a cell made of?

- **Chemical composition** - by weight
  - **70%** water
  - **7%** small molecules
    - ➢ salts
    - ➢ lipids
    - ➢ amino acids
    - ➢ nucleotides
  - **23%** macromolecules
    - ➢ proteins
    - ➢ polysaccharides
    - ➢ lipids

Types of cells: Prokaryotes vs. Eukaryotes



A prokaryotic cell

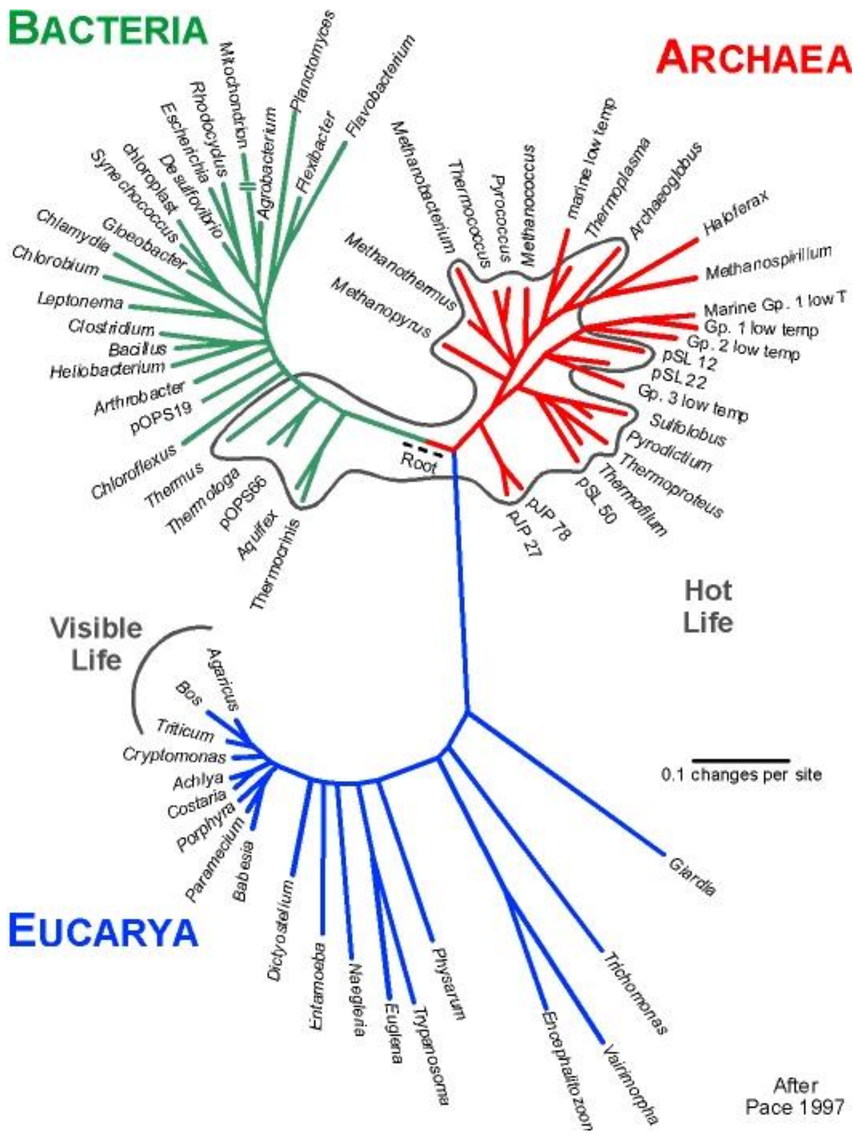


A eukaryotic cell

Prokaryotes vs. Eukaryotes

Prokaryotes	Eukaryotes
Single cell	Single or multi cell
No nucleus	Nucleus
No organelles	Organelles
One piece of circular DNA	Chromosomes
No mRNA post transcriptional modification	Exons/Introns splicing

Kingdoms of life



- According to the most recent evidence, there are three main branches in the tree of life.
- Prokaryotes include Archaea (“ancient ones”) and bacteria.
- Eukaryotes are kingdom Eukarya which includes plants, animals, fungi and certain algae.

From Pace NR (1997)
Science 276:734

Terminology

- **Genome**: an organism's genetic material
  - The **genome** is an organism's complete set of DNA
    - A bacteria contains about 600,000 DNA base pairs (bp)
    - Human and mouse genomes have some 3 billion bp
  - The human genome has 24 *distinct* **chromosomes**
    - 22 autosomal chromosome pairs, plus sex-determining X and Y chromosomes
    - Each chromosome contains many **genes**
- **Gene**: a discrete unit of hereditary information located on the chromosomes and consisting of DNA
  - Basic physical and functional units of heredity
  - Specific sequences of DNA bases that encode instructions on how to make **proteins**
- **Nucleotides** are biological molecules that form the building blocks of nucleic acids (DNA and RNA)

Terminology (continue)

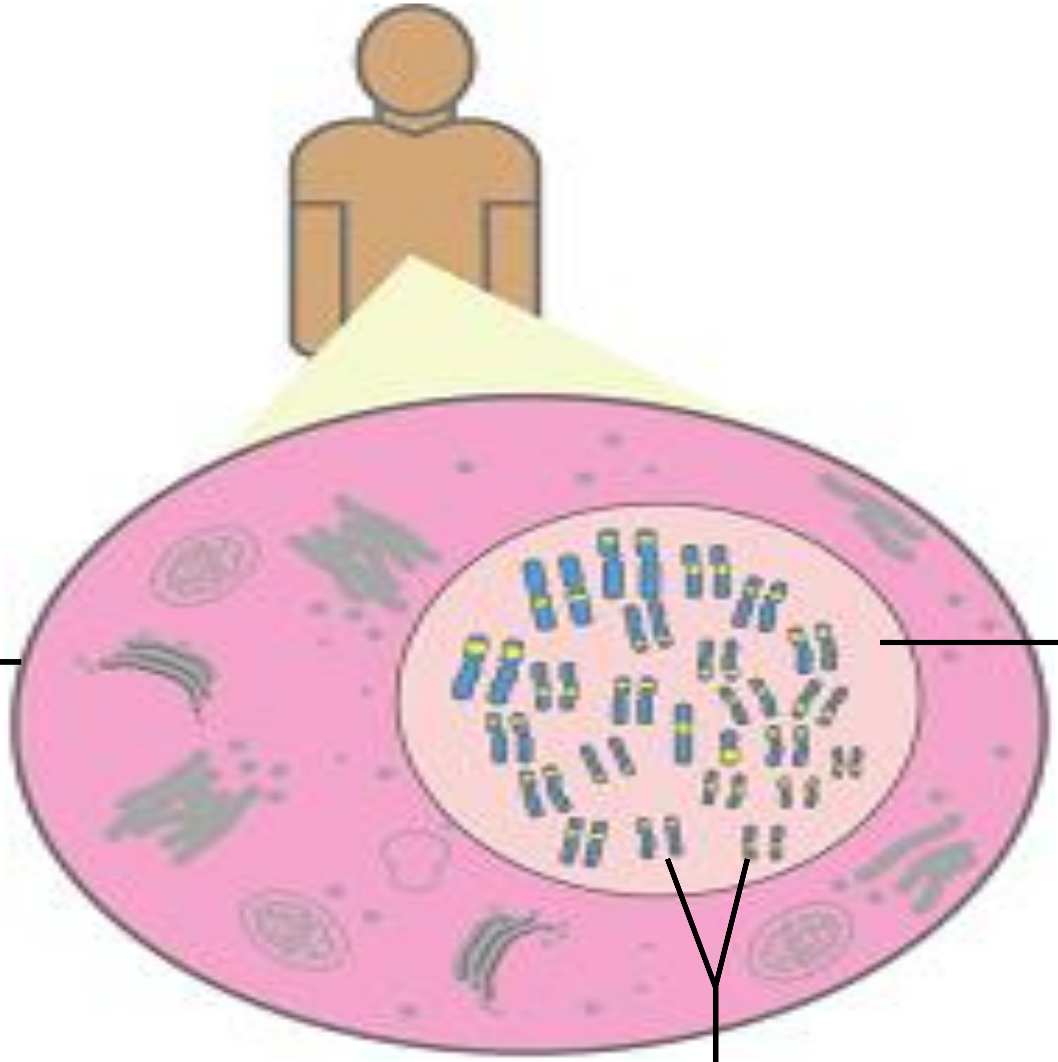
- **Proteins**
  - Make up the cellular structure
  - Large, complex molecules made up of smaller subunits called **amino acids**

- **Amino acids** are organic compounds that combine to form proteins

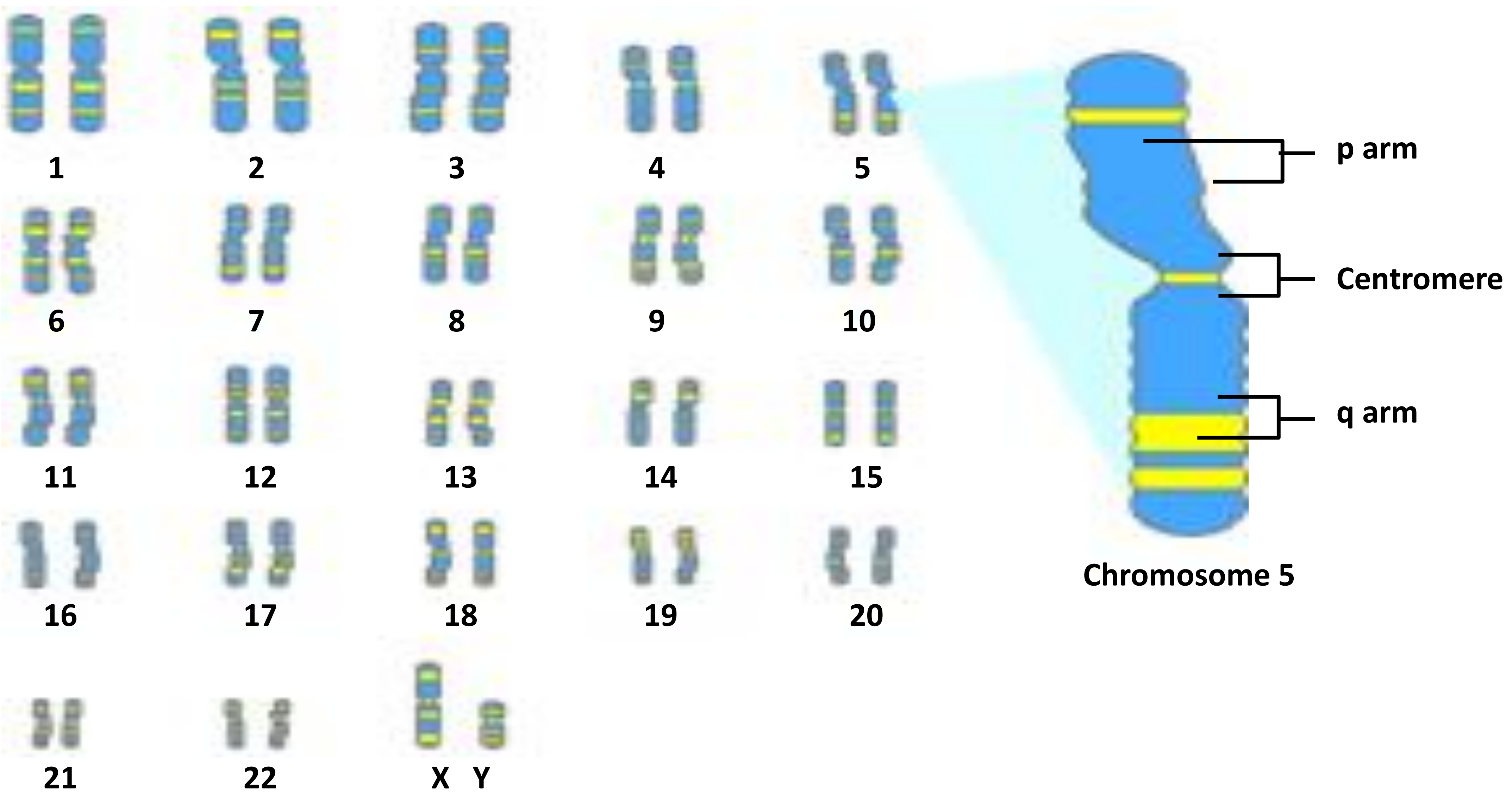
- **Genotype**: The genetic makeup of an organism

- **Phenotype**: the physical expressed traits of an organism

Genome



Chromosome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21
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X
Y
p arm
Centromere
q arm
Chromosome 5

Overview of organizations of life

- **Nucleus = library**
- **Chromosomes = bookshelves**
- **Genes = books**
- Almost every cell in an organism contains the same libraries and the same sets of books.
- Books represent almost all the information (DNA) that every cell in the body needs so it can grow and carry out its various functions.

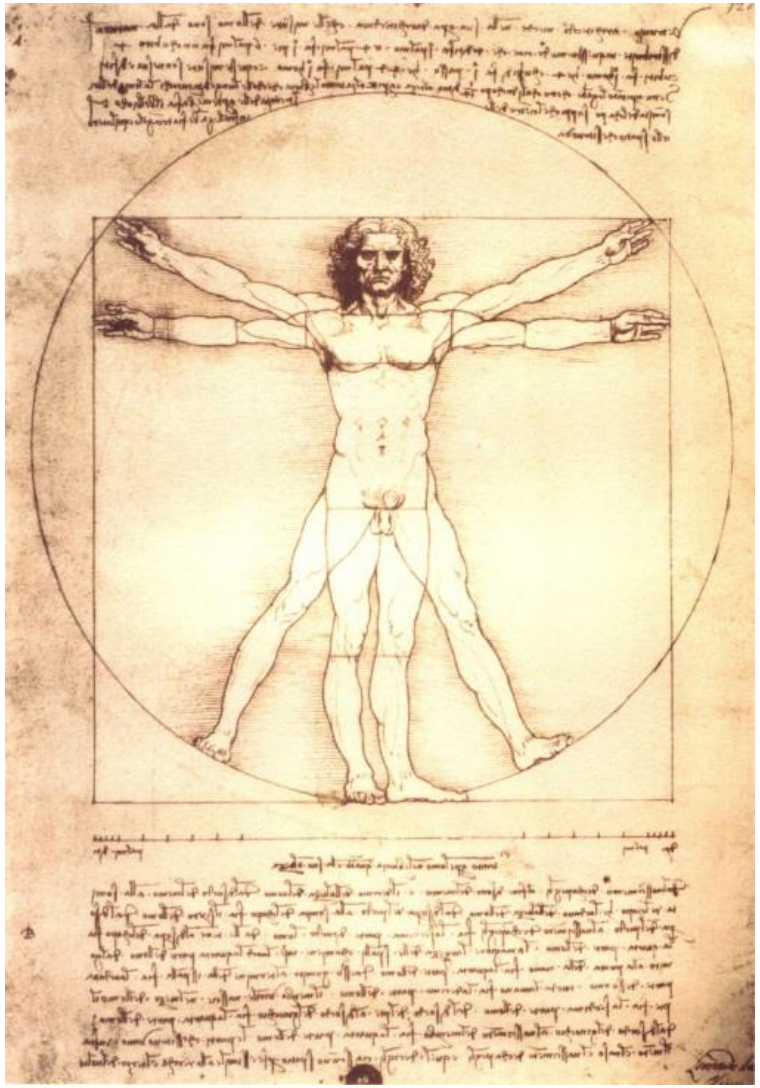
Cell information and machinery

- Cells store all information to replicate itself
  - Human genome is around 3 billion base pairs (bp) long
  - Almost every cell in human body contains the same set of genes
  - But not all genes are used or expressed by those cells

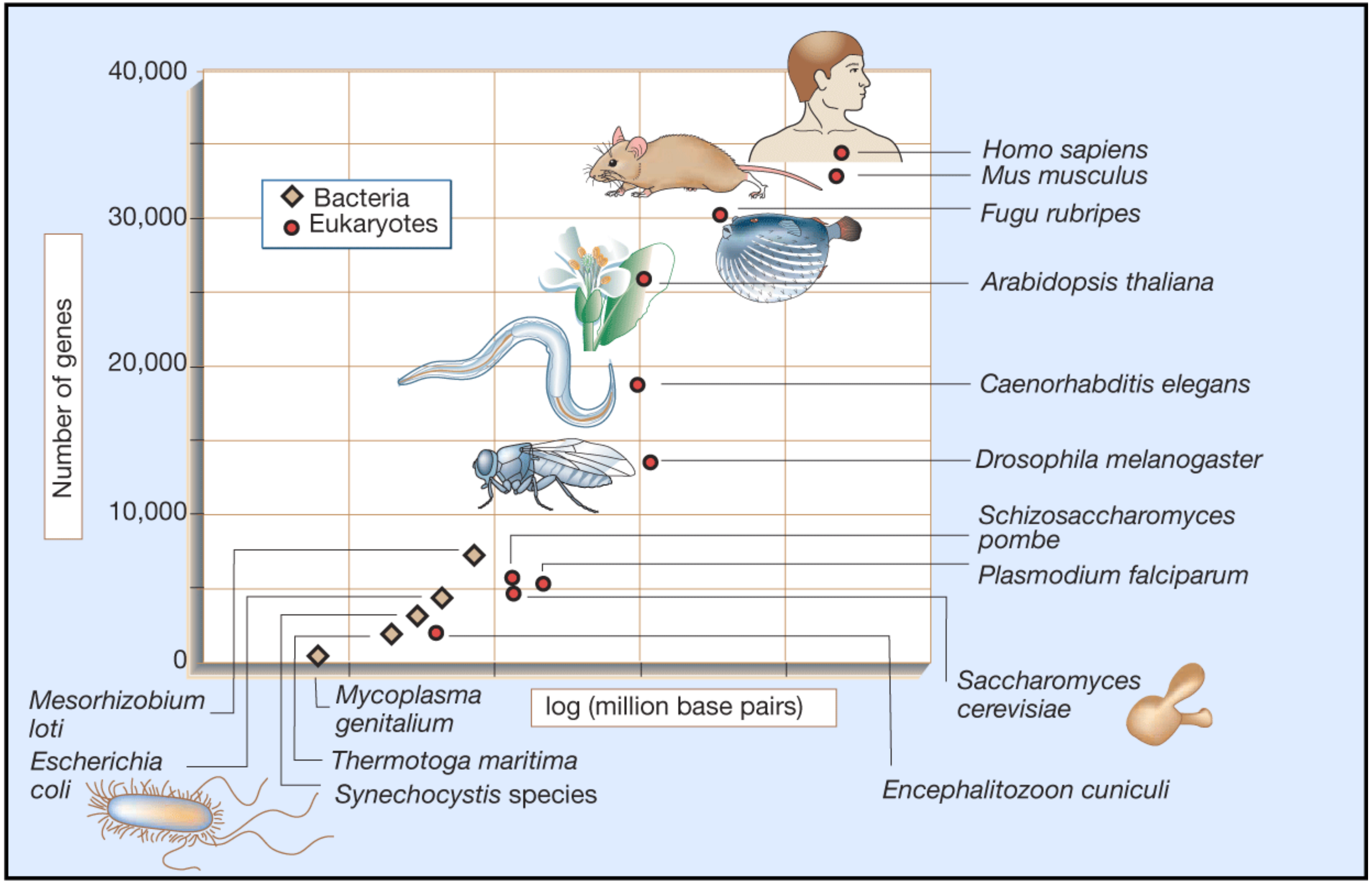
- Machinery:
  - Collect and manufacture components
  - Carry out replication
  - Kick-start its new offspring

Cells in the human body

- About 40-60 x 10^{12} cells in total
- Each cell
  - 23 pairs of chromosomes
  - 3.2 billion pairs of DNA bases
  - ~ 25,000 genes



Sizes of genomes and numbers of genes



The Central Dogma of Molecular Biology

All life depends on 3 critical molecules

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- Deoxyribonucleic acid (**DNA**)
  - Hold information on how cell works
- Ribonucleic acid (**RNA**)
  - Act to transfer short pieces of information to different parts of a cell
  - Provide templates to synthesize proteins
- **Protein**
  - Form enzymes that send signals to other cells and regulate gene activity
  - Form body’ s major components (e.g. hair, skin, etc.)

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Watson & Crick – *"…the secret of life"*

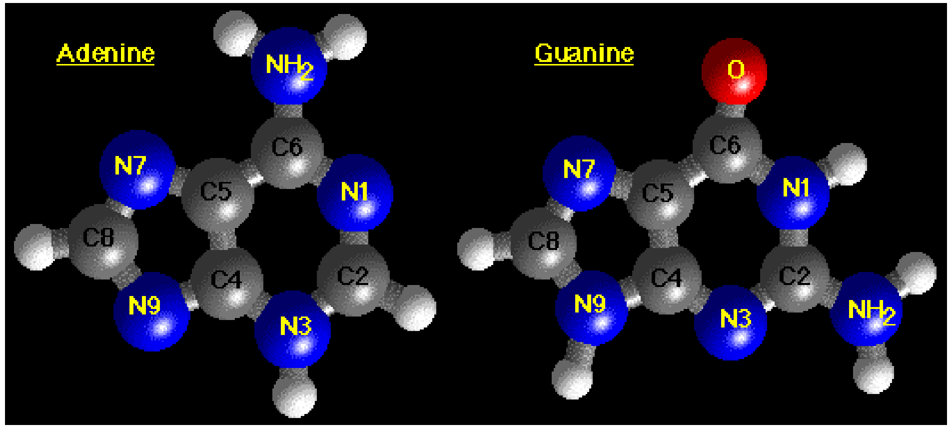
- J. Watson: a zoologist, F. Crick: a physicist
- "In 1947 Crick knew no biology and practically no organic chemistry or crystallography."
- Applying Chagraff' s rules and the X-ray image from Rosalind Franklin, they constructed a "tinkertoy" model showing the double helix
- Their 1953 *Nature* paper: "It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material."

Watson & Crick with DNA model

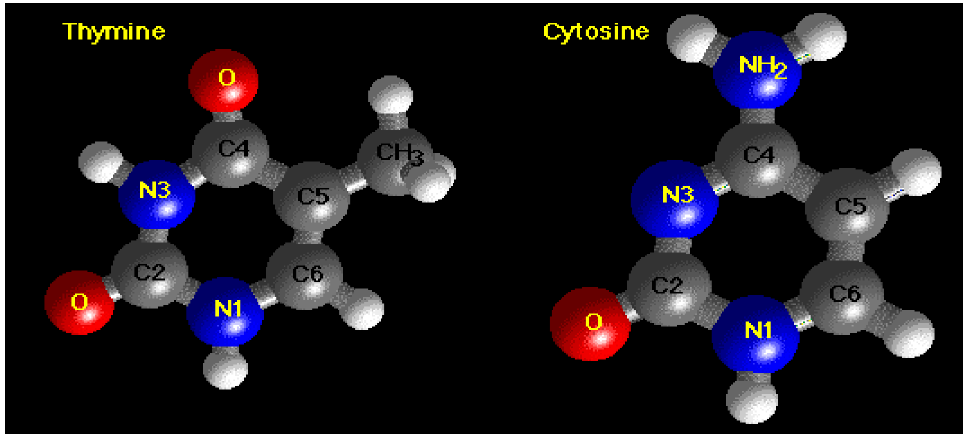
Rosalind Franklin with X-ray image of DNA

DNA base structure

- Structures of A & G (Purines)

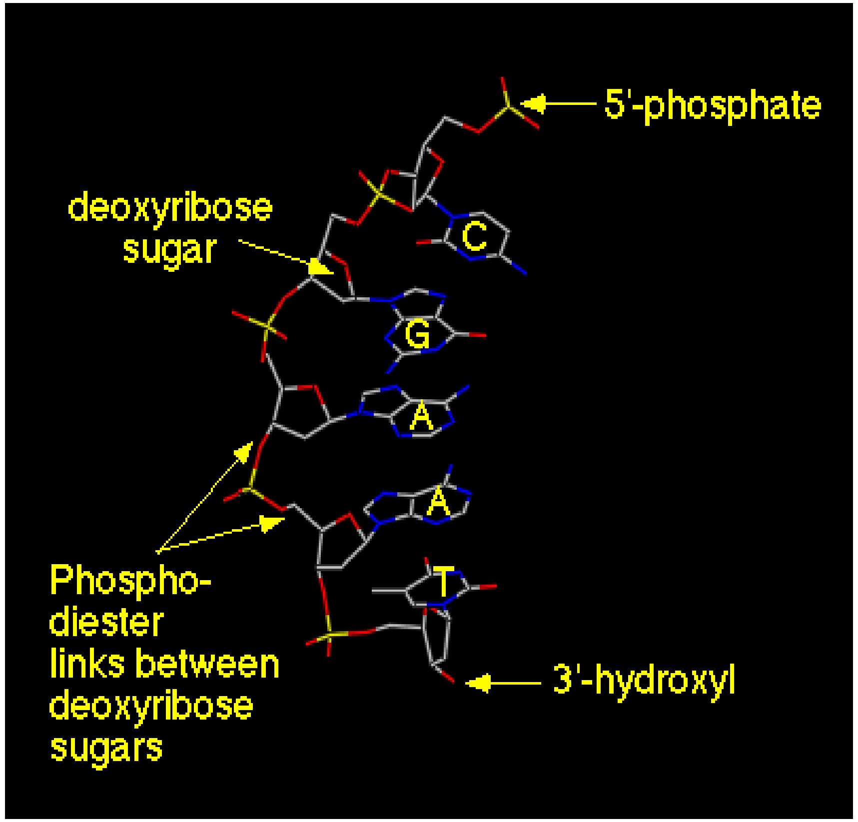


- Structures of T & C (Pyrimidines)



DNA backbone

- Alternating backbone of deoxyribose and phosphodiester groups
- Chain has a direction (known as **polarity**), 5'- to 3'- from top to bottom
- Oxygens (red atoms) of phosphates are polar and negatively charged
- A, G, C, and T bases extend away from chain, and stack on-top each other
- Bases are **hydrophobic**



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Nucleic acid

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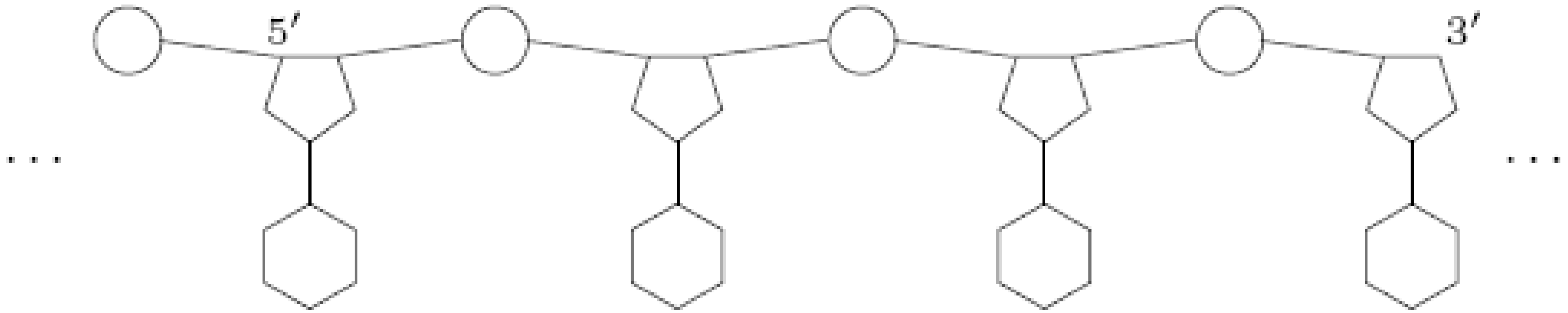
- Nucleic acids are responsible for encoding and storing the heredity information
- Schematic view of a nucleotide chain in the 5′ - 3′ direction
- Two different types of nucleic acids
  - Deoxyribonucleic acid (DNA)
  - Ribonucleic acid (RNA)



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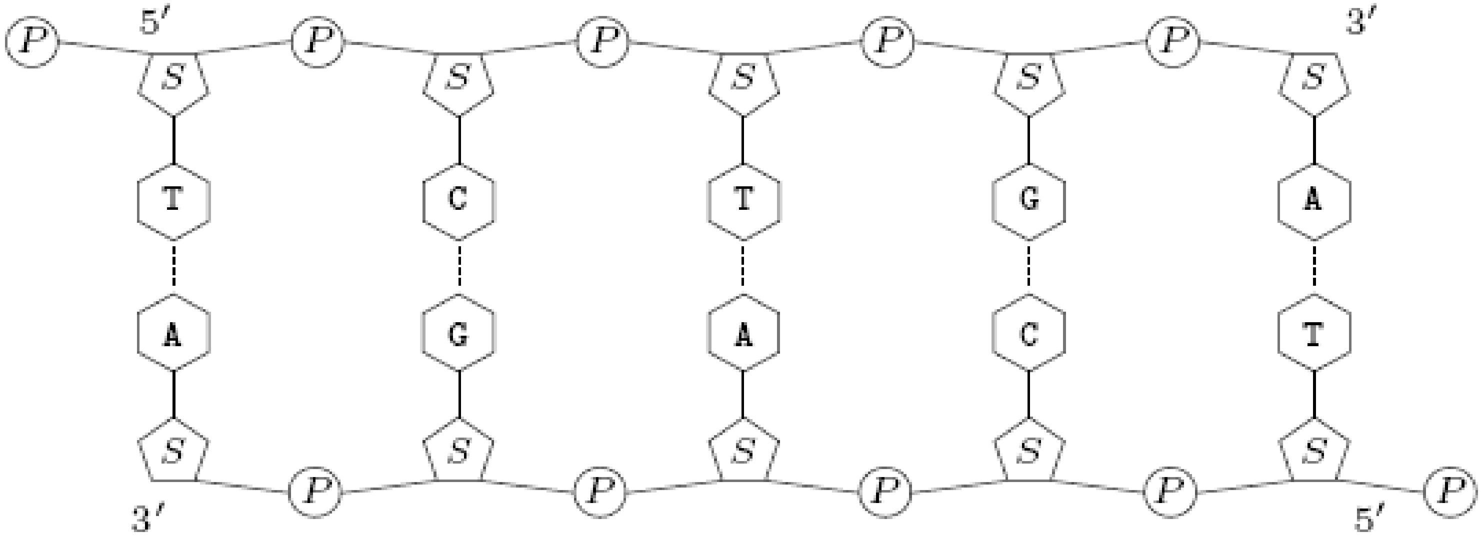
Example

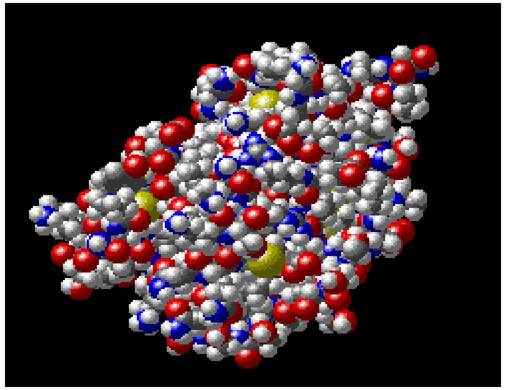
- Let s = TCTGA be the string representation of a DNA strand
- Then s~ = TCAGA is the reverse complement of s,
  - Unit of measurement: bp (standing for base pair)
  - By convention, DNA sequences are written in a 5' -to-3' direction

```
s   : 5' : : : TCTGA : : : 3'
s~  : 3' : : : AGACT : : : 5'
```



Genetic information

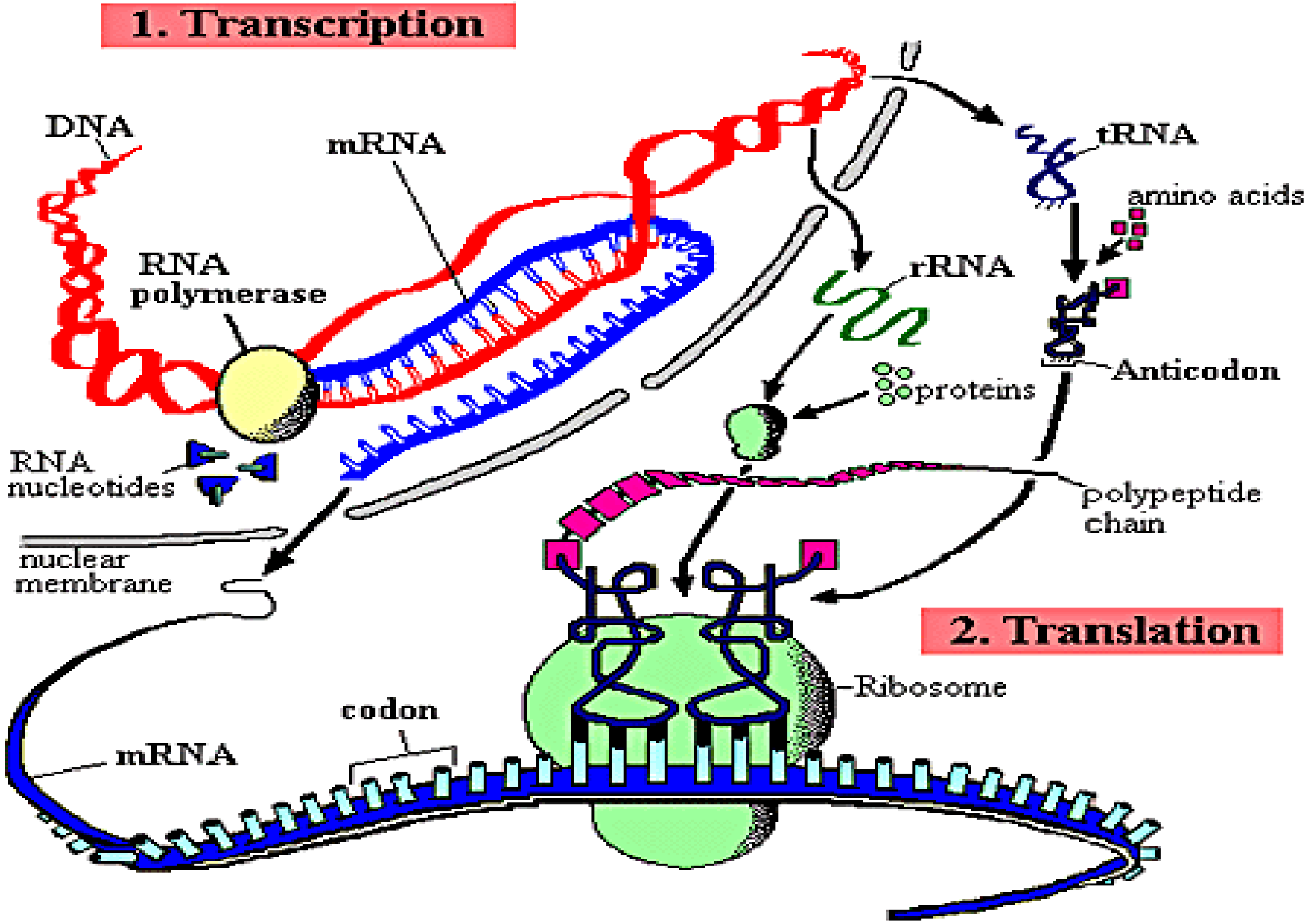
DNA $\xrightarrow{\text{transcription}}$ RNA $\xrightarrow{\text{translation}}$ Protein $\longrightarrow$ Phenotype



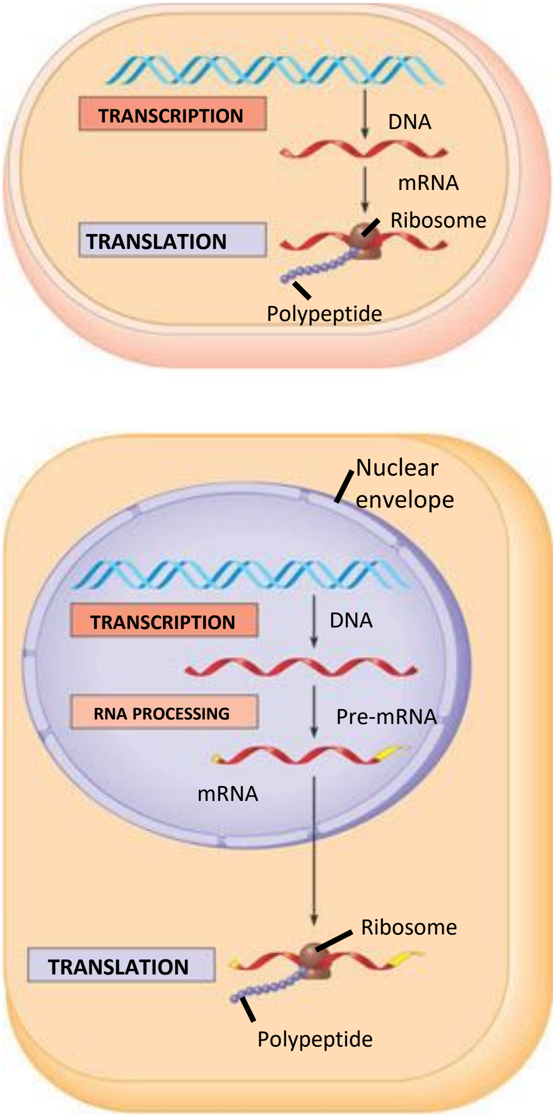
The Central Dogma of molecular biology

- **A fundamental principle of molecular biology**

  - Genetic information flows
    from DNA
    to RNA
    to protein



Transcription and translation in the flow of genetic information



(a) **Prokaryotic cell.** In a cell lacking a nucleus, mRNA produced by transcription is immediately translated without additional processing.

(b) **Eukaryotic cell.** The nucleus provides a separate compartment for transcription. The original RNA transcript, called pre-mRNA, is processed in various ways before leaving the nucleus as mRNA.

Recap

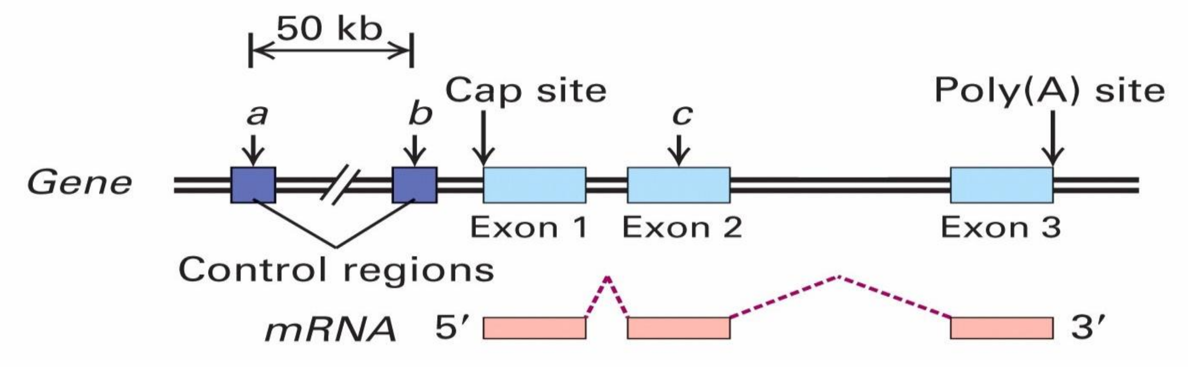
- In this lecture, we have learned:
  - The historical and intellectual roots of molecular biology
  - What is life made of?
  - The Central Dogma of Molecular Biology
  - How are RNA produced by transcription?
  - How are proteins produced by translation?

References

- E. Lander and M. Waterman (1995). "**The Secrets of Life: A Mathematician's Introduction to Molecular Biology**", Chapter 1 of the National Research Council report entitled "*Calculating the Secrets of Life: Contributions of the Mathematical Sciences to Molecular Biology*", National Academy Press, Washington D.C., 1995.
- N. Jones and P. Pevzner (2004). "**Molecular Biology Primer**", Chapter 3 of their book "*An Introduction to Bioinformatics Algorithms*", MIT Press, 2004.
- B. Alberts, et al. "**Molecular Biology of the Cell**", 7^{th} Edition, New York: W. W. Norton & Company, 2022.

Appendix A: Transcriptional Regulation

Structure of a gene



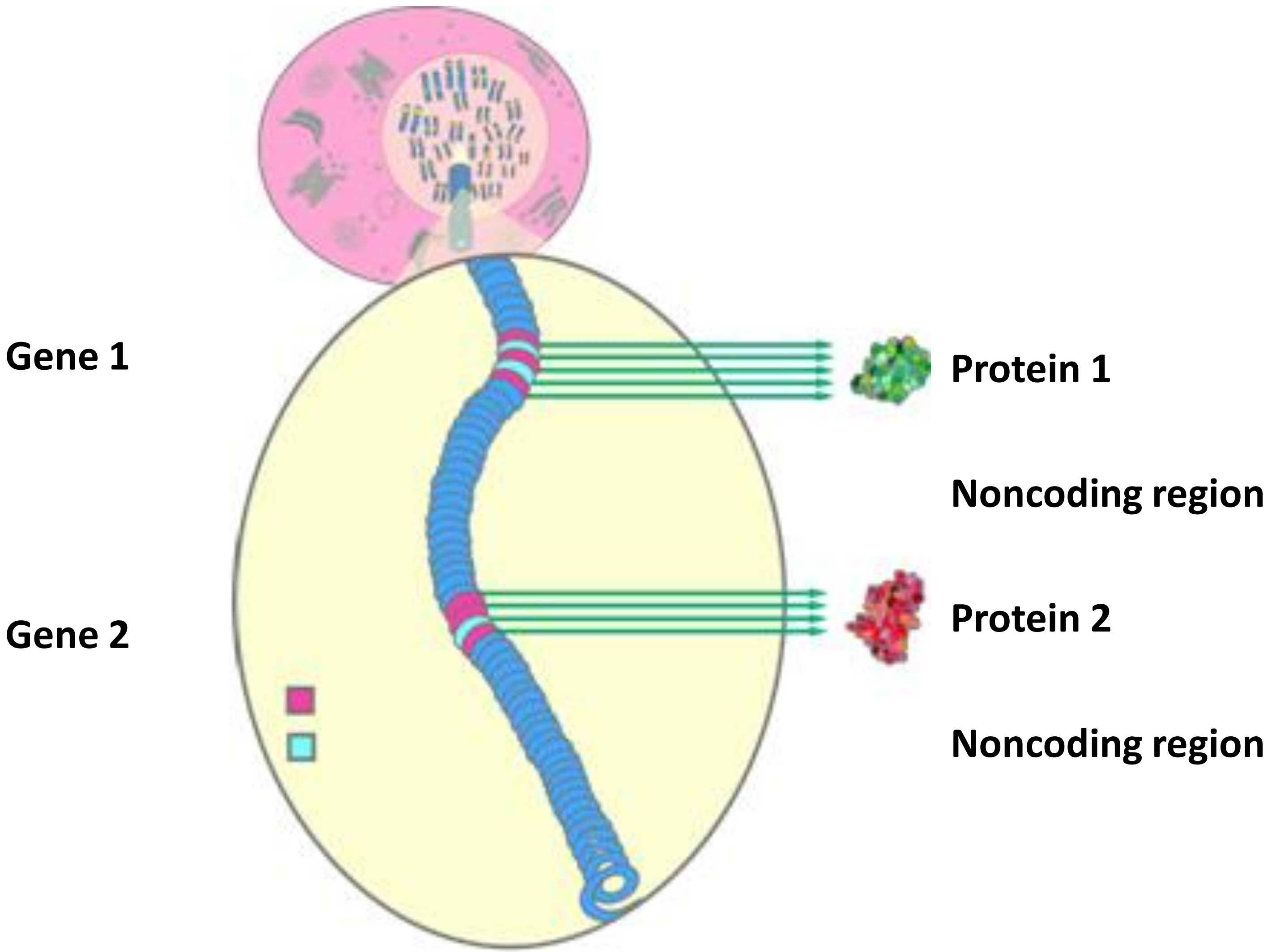
- Regulatory regions: up to 50 kb upstream of +1 site

- Exons: Protein coding and untranslated regions (UTR)
  1 to 178 exons per gene (mean 8.8)
  8 bp to 17 kb per exon (mean 145 bp)

- Introns: Splice acceptor and donor sites, junk DNA
  Average 1 kb – 50 kb per intron

- Gene size: Largest – 2.4 Mb (Dystrophin). Mean – 27 kb.

Noncoding regions



mRNA transcription

- **mRNA (messenger RNA)** carries the information encoded in DNA out of the nucleus to the protein assembly machinery, called the ribosome, in the cytoplasm.
- **mRNA transcription** - the synthesis of mRNA from a DNA
  - coding portions of a gene, exons, interrupted by introns
  - editing process removes the introns, to make a **mature mRNA**

mRNA transcription process

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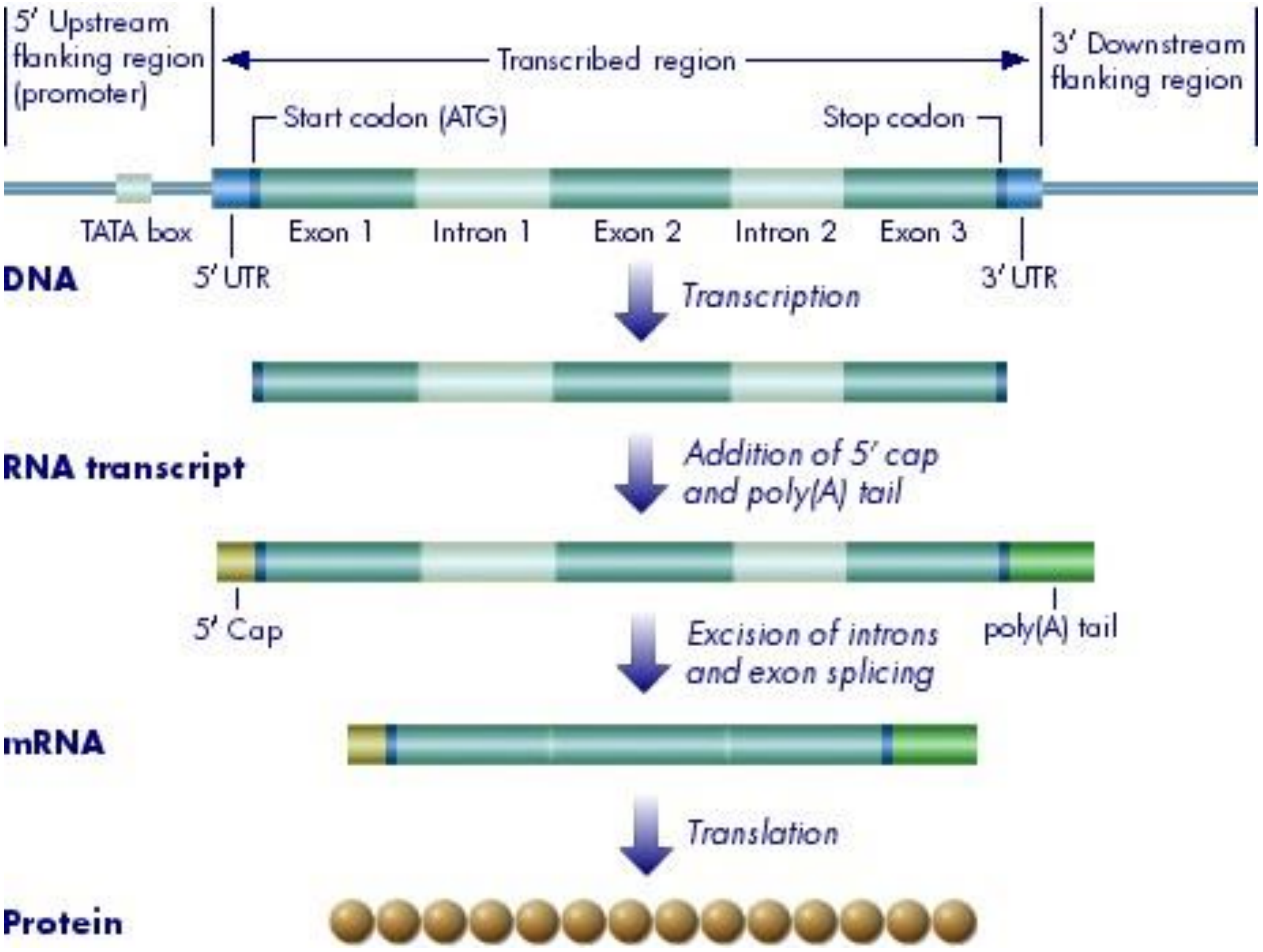
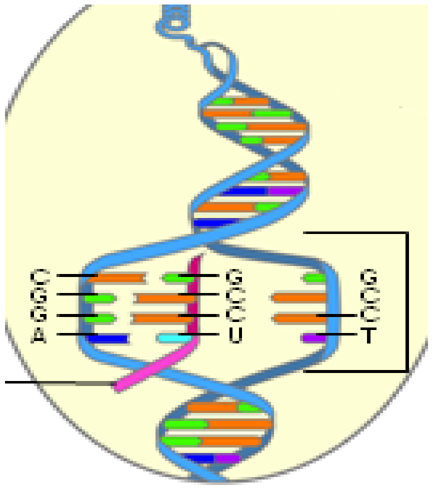
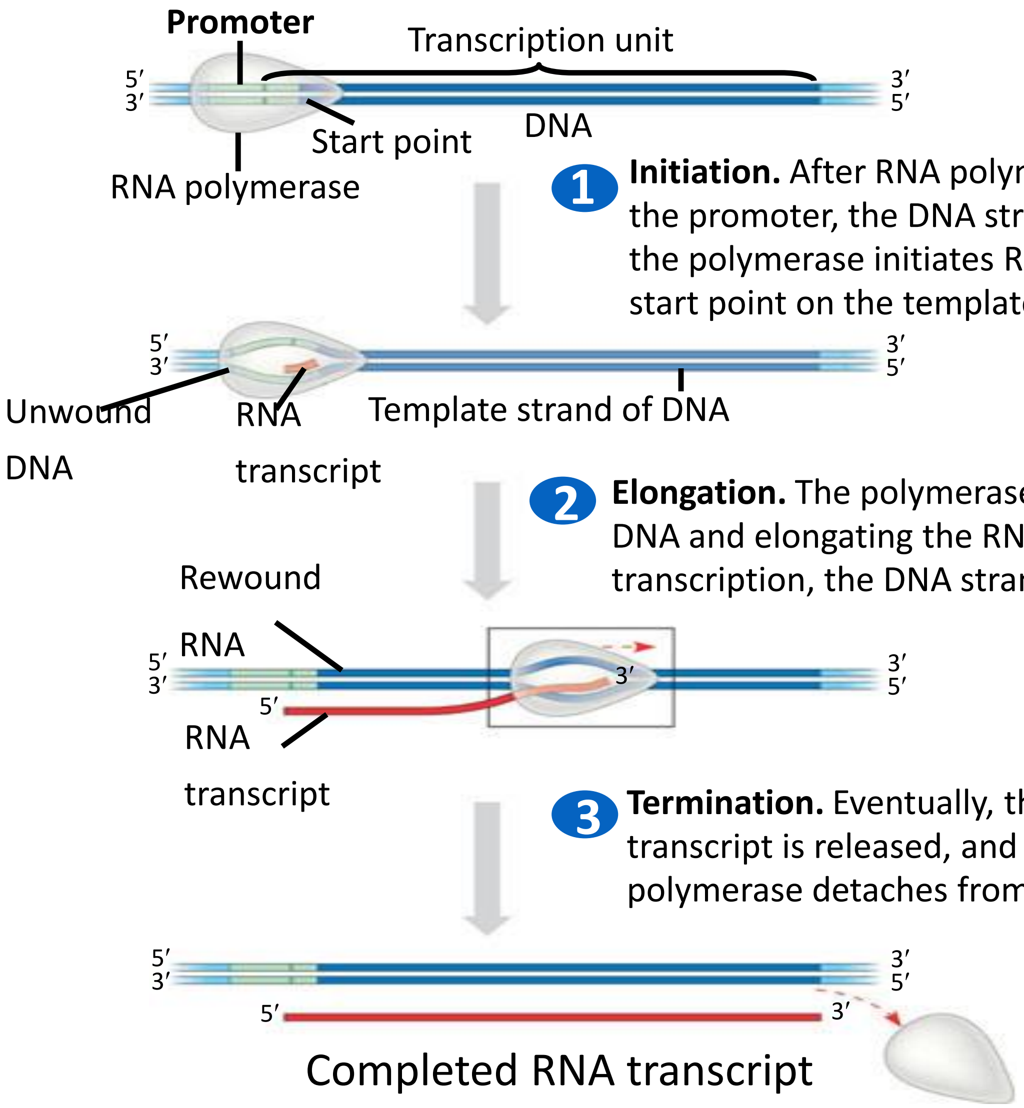




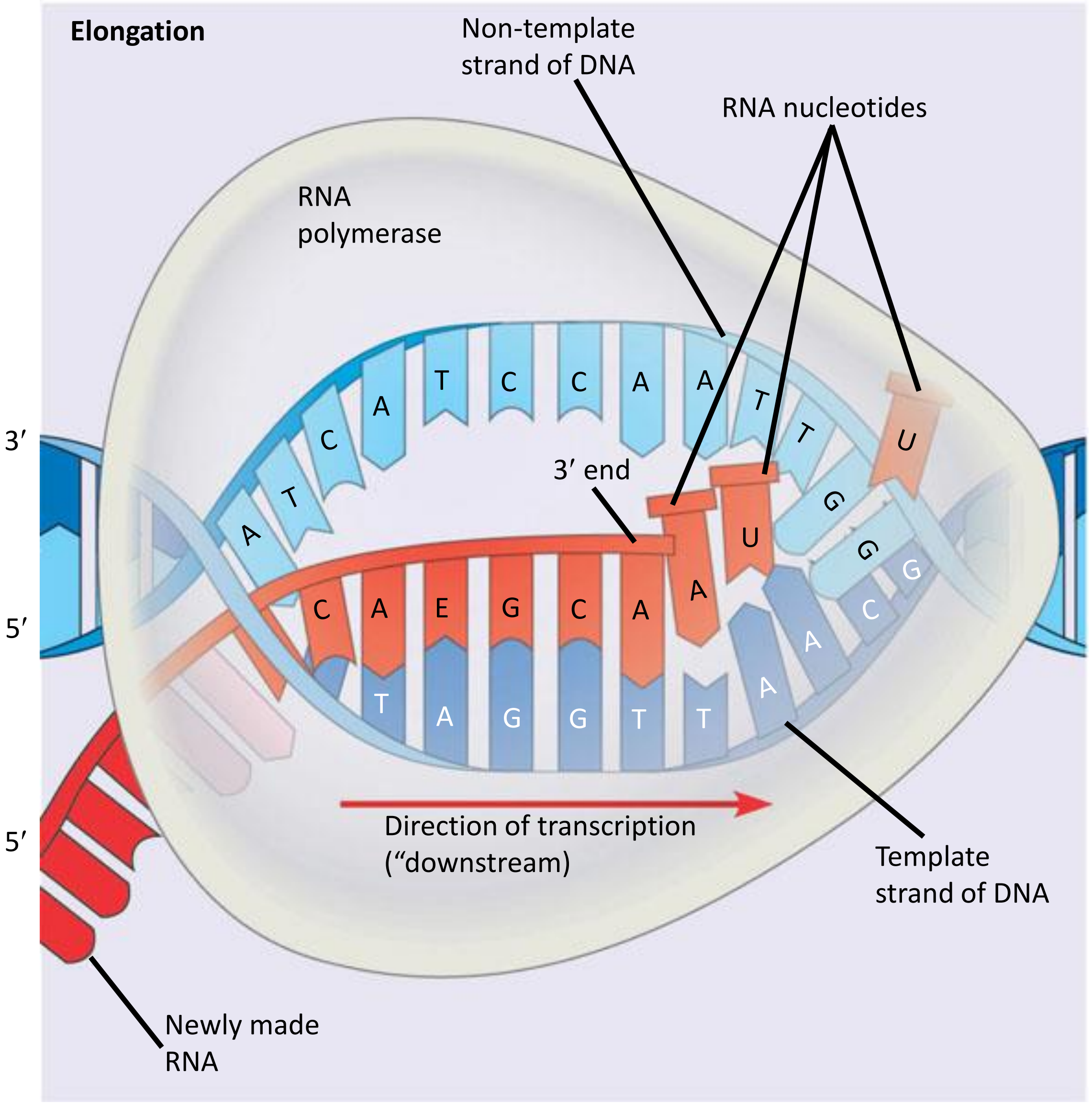
Initiation, elongation, and termination



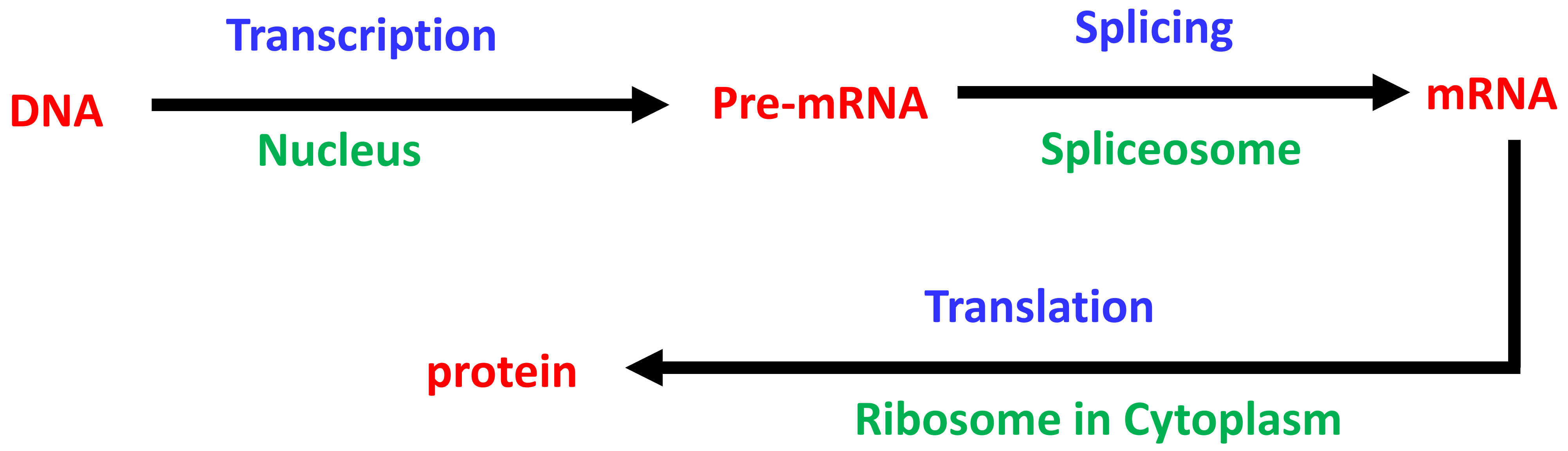
1 **Initiation.** After RNA polymerase binds to the promoter, the DNA strands unwind, and the polymerase initiates RNA synthesis at the start point on the template strand.

2 **Elongation.** The polymerase moves downstream, unwinding the DNA and elongating the RNA transcript 5′ → 3′. In the wake of transcription, the DNA strands re-form a double helix.

3 **Termination.** Eventually, the RNA transcript is released, and the polymerase detaches from the DNA.


Elongation
Non-template strand of DNA
RNA nucleotides
RNA polymerase
3′
5′
5′
3′ end
A T C A T C C A A T T G G
C A E G C A A U U
T A G G T T A A C G
Direction of transcription (“downstream)
Template strand of DNA
Newly made RNA

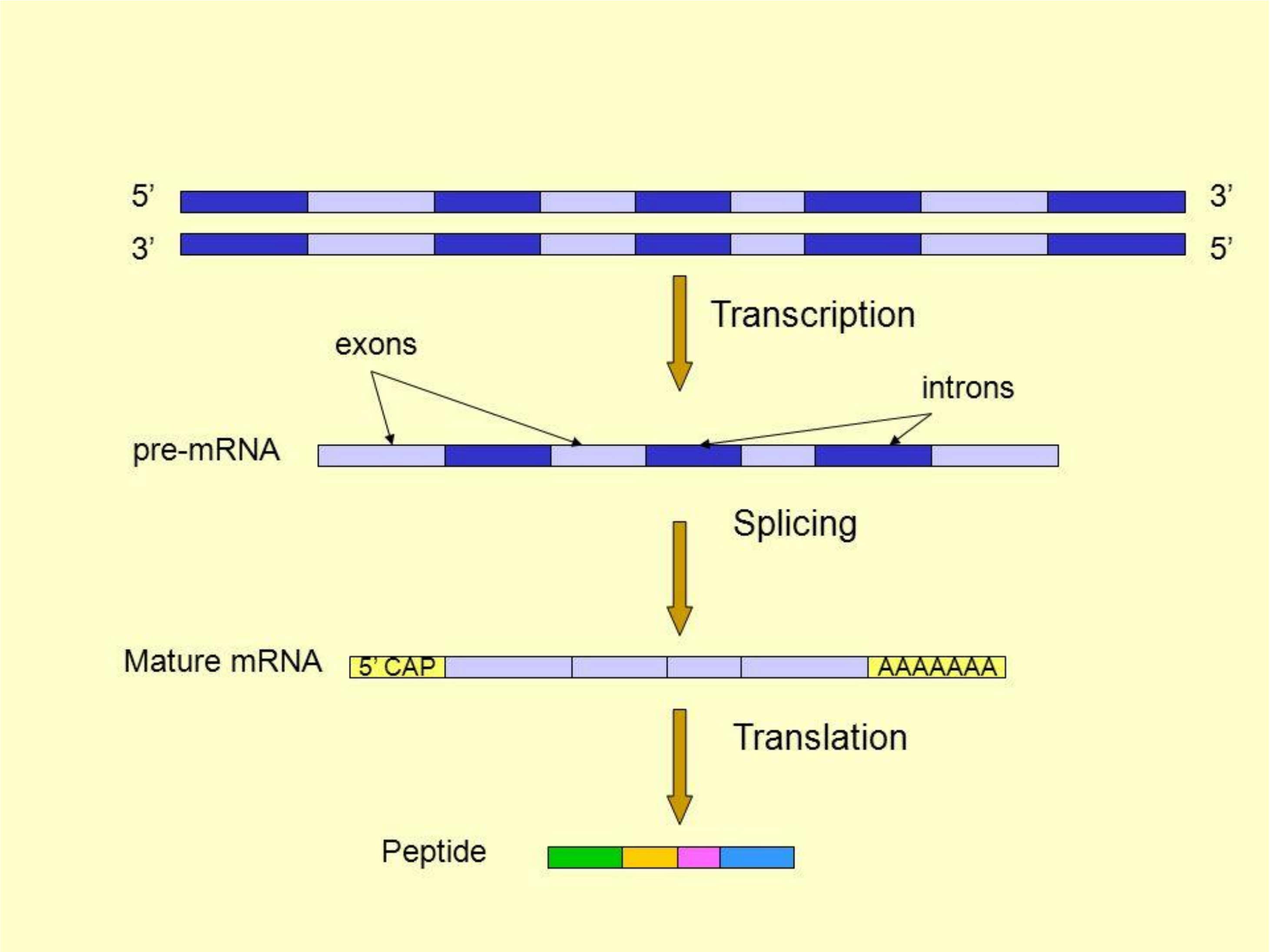
Central dogma revisited



- **Base Pairing Rule**: A and T (or U) is held together by 2 hydrogen bonds and G and C is held together by 3 hydrogen bonds
- **Note**: Some mRNA stays as RNA (e.g. tRNA, rRNA)

Splicing

- Splice variants create an enormous diversity of proteins
  - ~25,000 genes in humans give rise to 200,000 to 2,000,000 different proteins
  - Splice variants may have very diverse functions



Alternative splicing

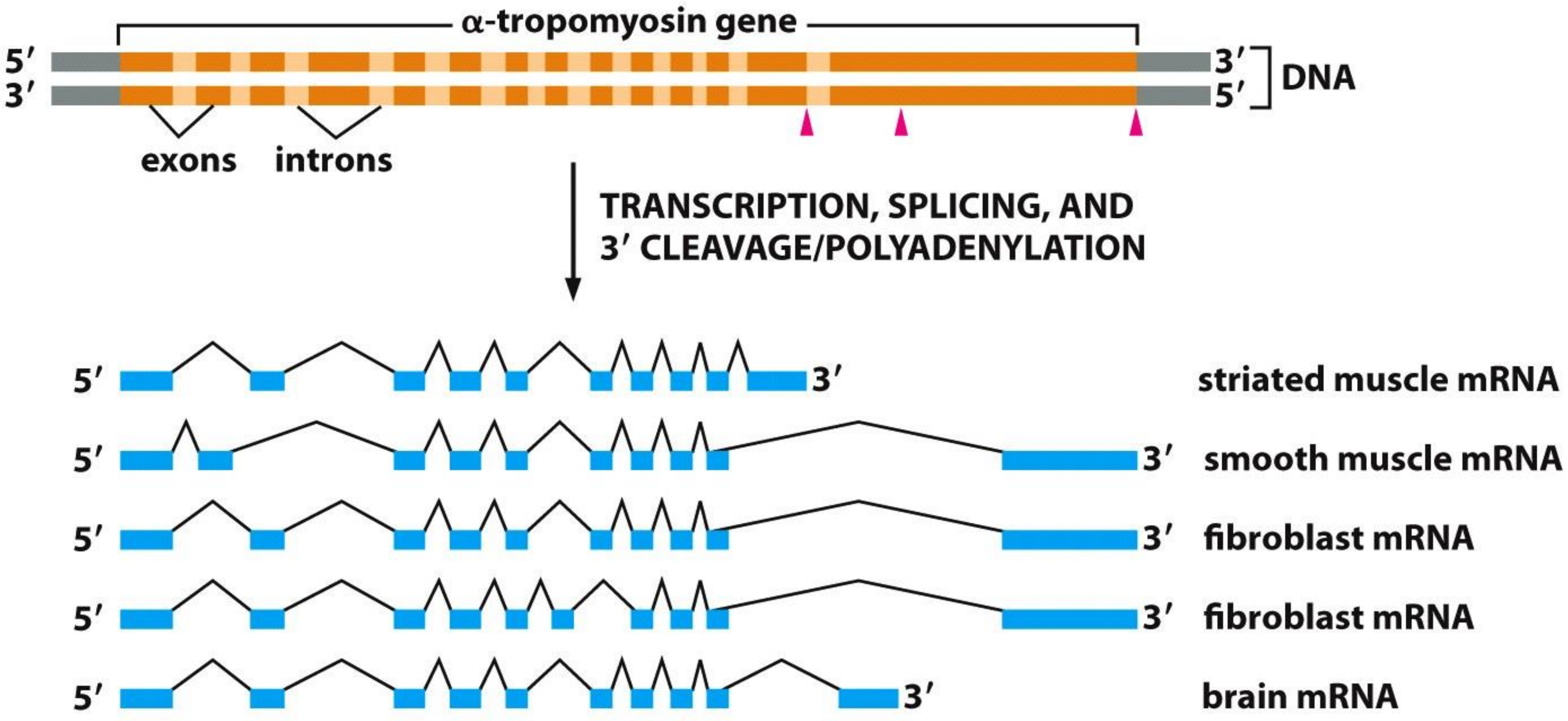


Figure 7-21 Essential Cell Biology 3/e (© Garland Science 2010)

Appendix B: Protein Translation

Protein translation

- DNA - carrier of genetic information
- Protein - do the bulk of the work in the cell
  - A long chain containing as many as 20 different kinds of amino acids
  - The order & number of amino acids determine the shape and function of the protein
- Each cell contains thousands of different proteins
- The ribosomal complex builds a protein one amino acid at a time, with the order of amino acids determined precisely by the order of the **codons** in the mRNA

Translation: the basic concept

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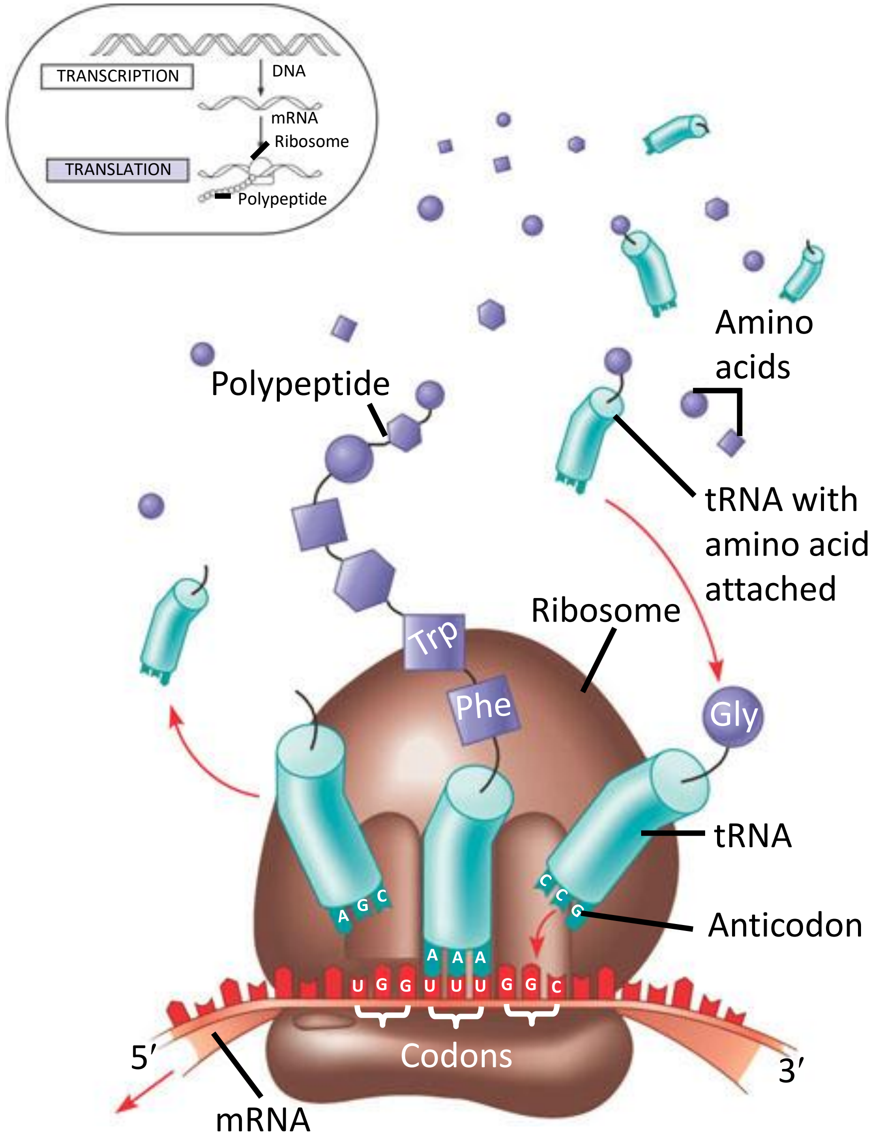

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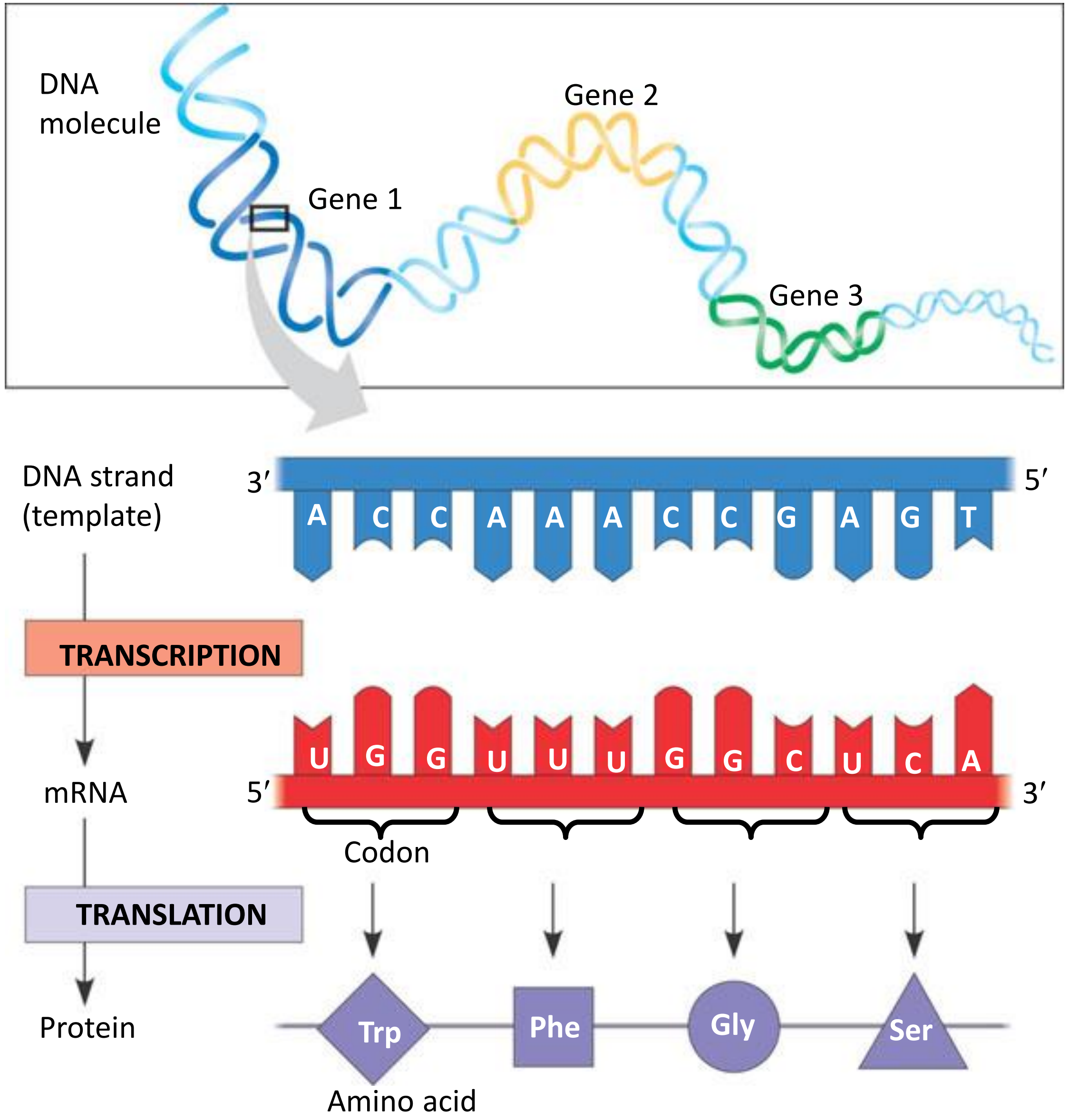
Terminology

- **Codon**: The sequence of 3 nucleotides in DNA/RNA that encodes for a specific amino acid
- **mRNA (messenger RNA)**: A ribonucleic acid whose sequence is complementary to that of a protein-coding gene in DNA
- **Ribosome**: The organelle that synthesizes polypeptides under the direction of mRNA
- **rRNA (ribosomal RNA)**: The RNA molecules that constitute the bulk of the ribosome, provide structural scaffolding for the ribosome and catalyze peptide bond formation
- **tRNA (transfer RNA)**: The small L-shaped RNAs that deliver specific amino acids to ribosomes according to the sequence of a bound mRNA

The triplet code



64(=4^3) RNA triplet codons and their corresponding amino acids

		2nd base			
		U	C	A	G
1st base	U	UUU (Phe/F)Phenylalanine UUC (Phe/F)Phenylalanine UUA (Leu/L)Leucine UUG (Leu/L)Leucine	UCU (Ser/S)Serine UCC (Ser/S)Serine UCA (Ser/S)Serine UCG (Ser/S)Serine	UAU (Tyr/Y)Tyrosine UAC (Tyr/Y)Tyrosine UAA Ochre (*Stop*) UAG Amber (*Stop*)	UGU (Cys/C)Cysteine UGC (Cys/C)Cysteine UGA Opal (*Stop*) UGG (Trp/W)Tryptophan
	C	CUU (Leu/L)Leucine CUC (Leu/L)Leucine CUA (Leu/L)Leucine CUG (Leu/L)Leucine	CCU (Pro/P)Proline CCC (Pro/P)Proline CCA (Pro/P)Proline CCG (Pro/P)Proline	CAU (His/H)Histidine CAC (His/H)Histidine CAA (Gln/Q)Glutamine CAG (Gln/Q)Glutamine	CGU (Arg/R)Arginine CGC (Arg/R)Arginine CGA (Arg/R)Arginine CGG (Arg/R)Arginine
	A	AUU (Ile/I)Isoleucine AUC (Ile/I)Isoleucine AUA (Ile/I)Isoleucine AUG (Met/M)Methionine, *Start*[1]	ACU (Thr/T)Threonine ACC (Thr/T)Threonine ACA (Thr/T)Threonine ACG (Thr/T)Threonine	AAU (Asn/N)Asparagine AAC (Asn/N)Asparagine AAA (Lys/K)Lysine AAG (Lys/K)Lysine	AGU (Ser/S)Serine AGC (Ser/S)Serine AGA (Arg/R)Arginine AGG (Arg/R)Arginine
	G	GUU (Val/V)Valine GUC (Val/V)Valine GUA (Val/V)Valine GUG (Val/V)Valine	GCU (Ala/A)Alanine GCC (Ala/A)Alanine GCA (Ala/A)Alanine GCG (Ala/A)Alanine	GAU (Asp/D)Aspartic acid GAC (Asp/D)Aspartic acid GAA (Glu/E)Glutamic acid GAG (Glu/E)Glutamic acid	GGU (Gly/G)Glycine GGC (Gly/G)Glycine GGA (Gly/G)Glycine GGG (Gly/G)Glycine

Types of codons

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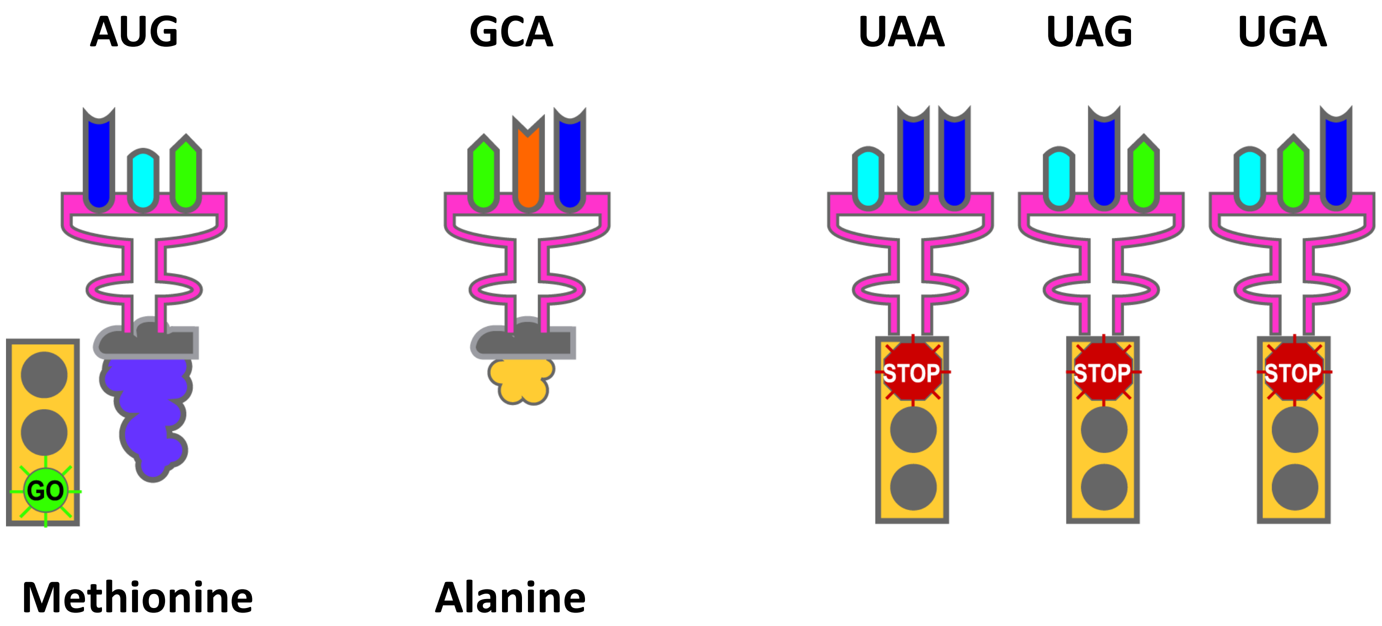

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Protein structure and function

- A protein is a **polypeptide**. However, it is a very difficult problem to understand the functions of a protein given only the polypeptide sequence
- **Protein folding** is an open problem. The 3D structure depends on many variables
- Improper folding of a protein is believed to be the cause of diseases, e.g. mad cow disease

Hierarchy of protein structures

```
DNA sequence      : 5′- ATG CCC TGC TTG GCC …- 3′
RNA sequence      : 5′- AUG CCC UGC UUG GCC …-3′
Protein sequence  :      M   P   C   L   A  …
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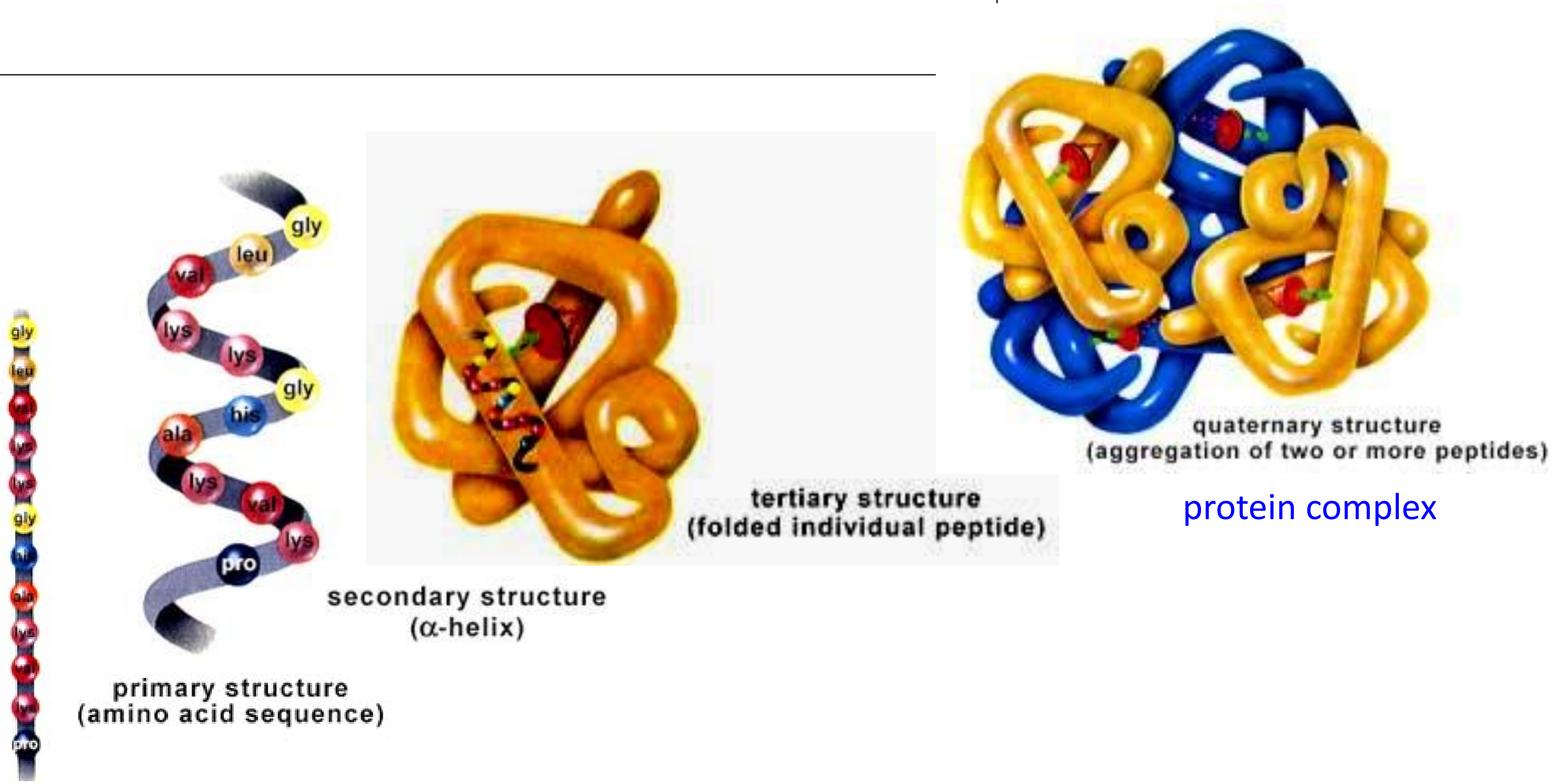
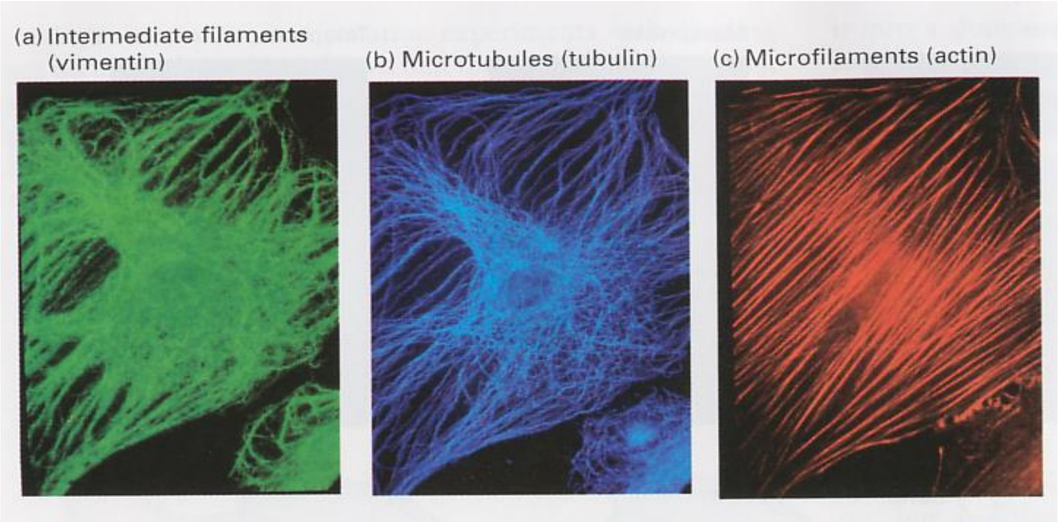
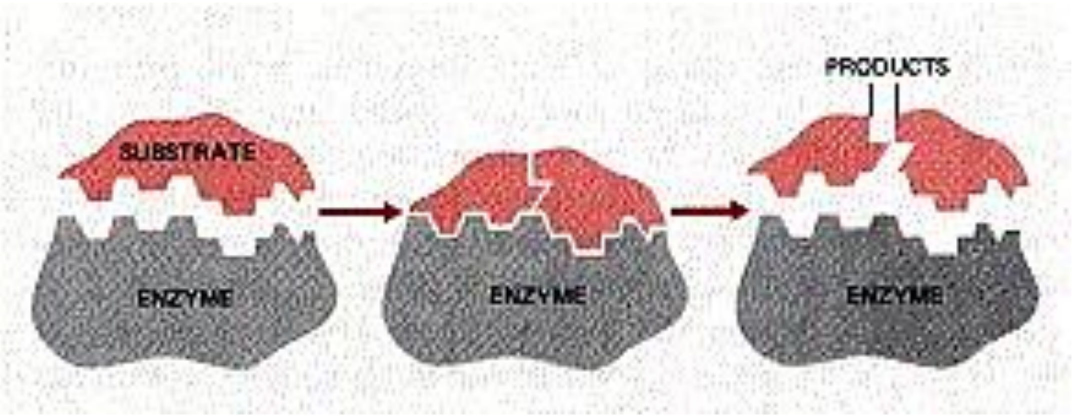


protein complex

Protein functions



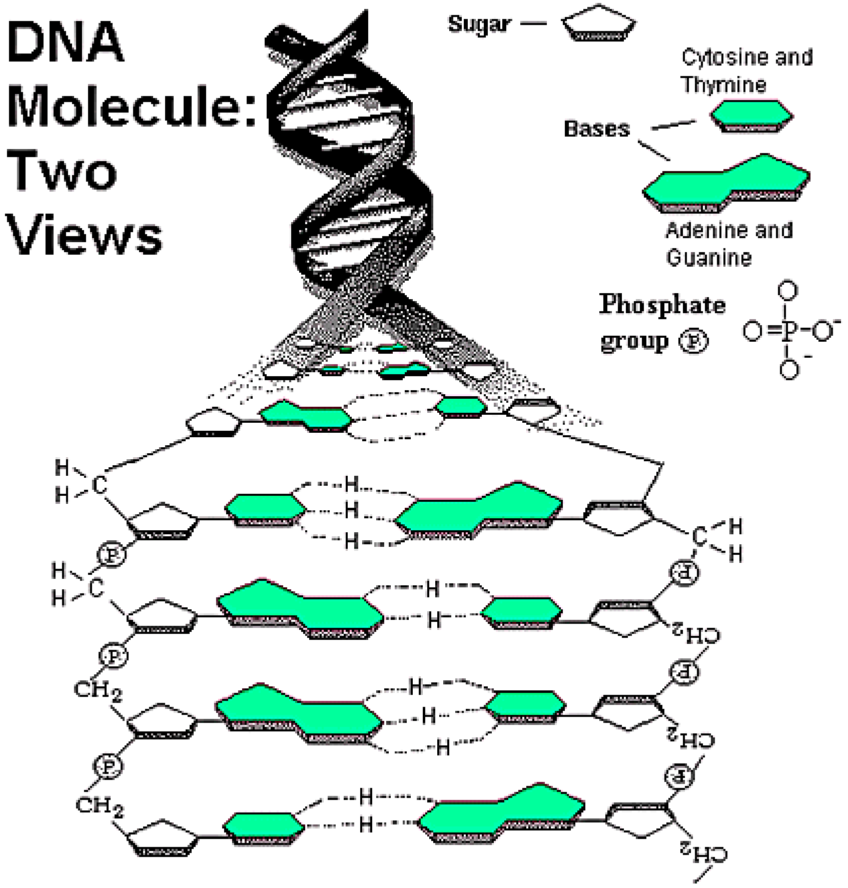
e.g. Structural role of proteins

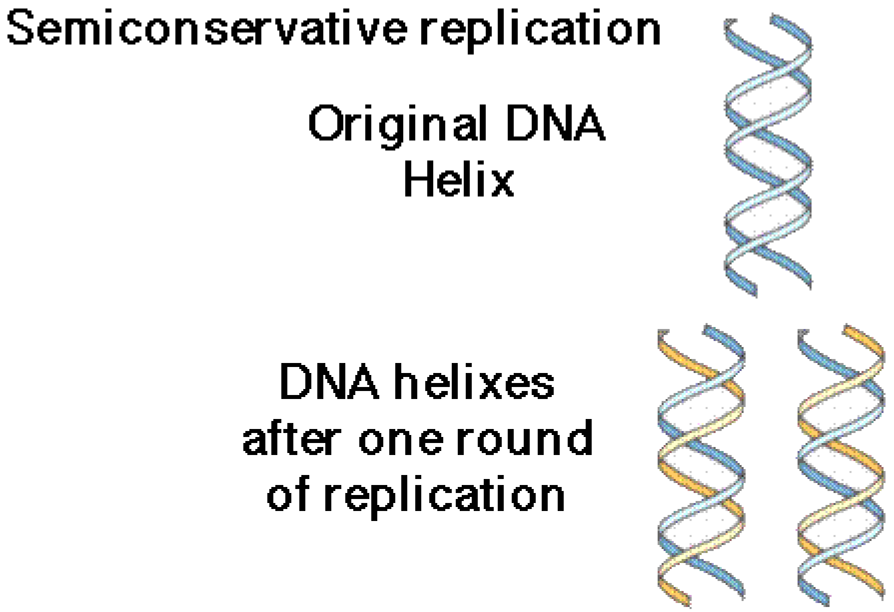


e.g. Proteins as enzymes (catalysts)

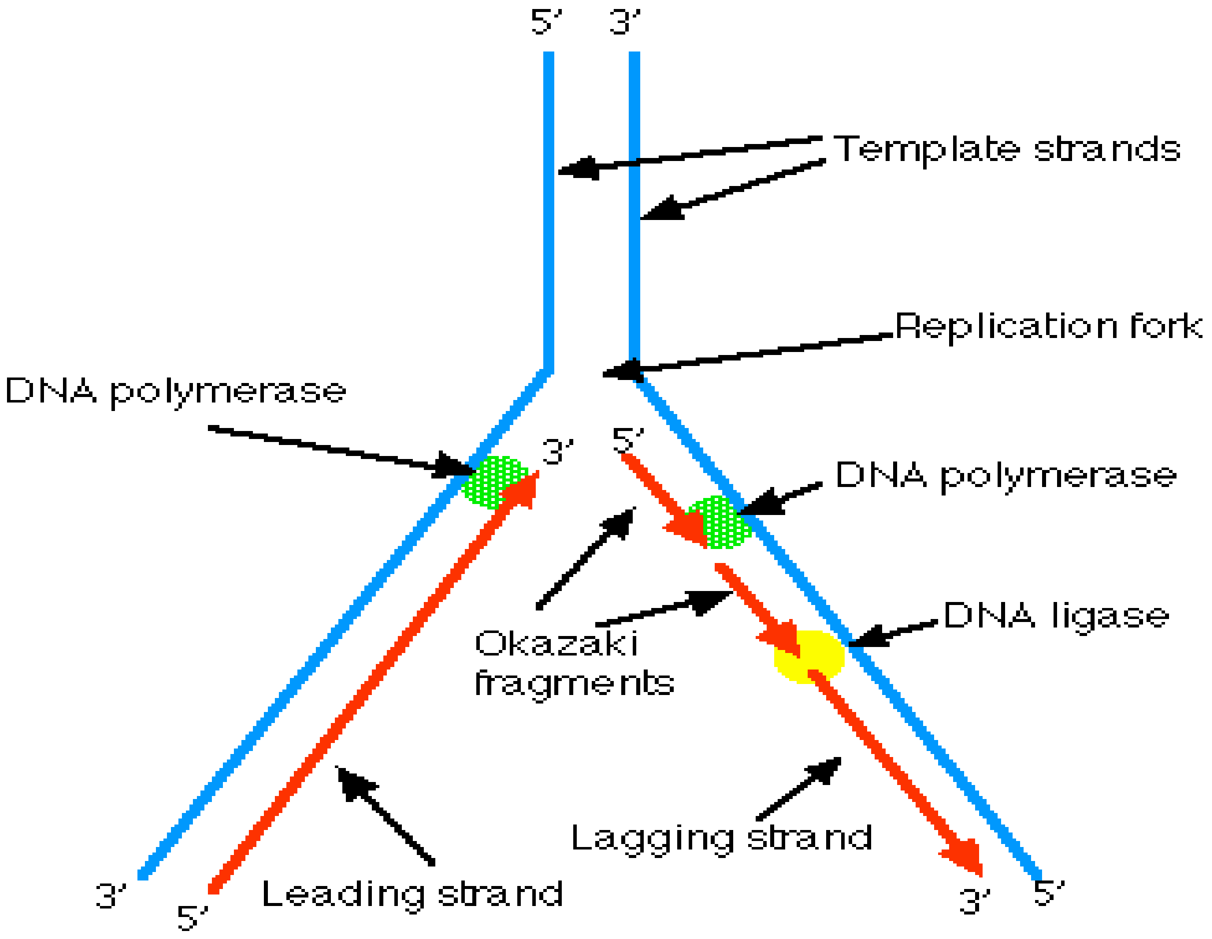
Appendix C: Replication and Reproduction

DNA replication: complementary strands



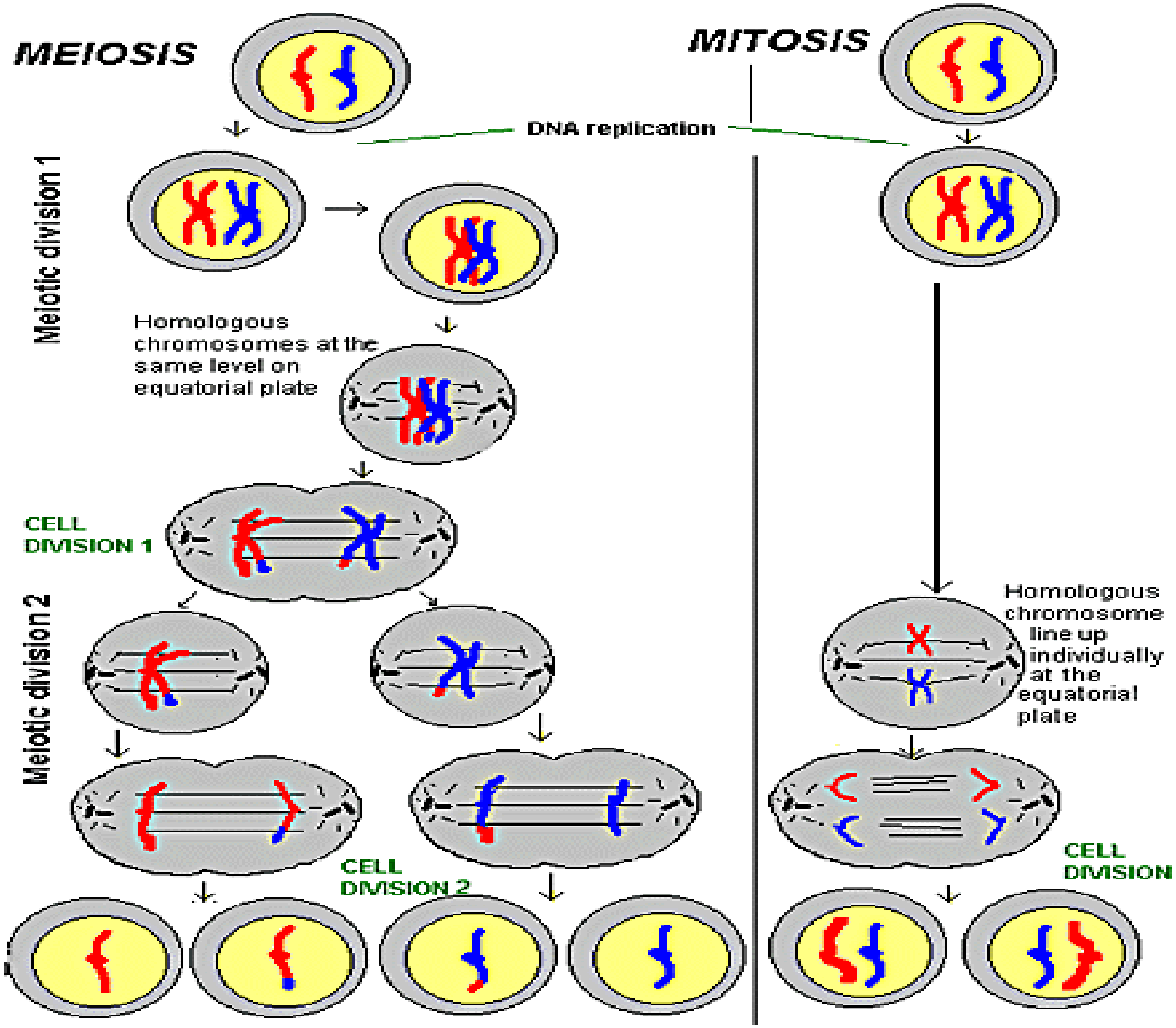


DNA replication process



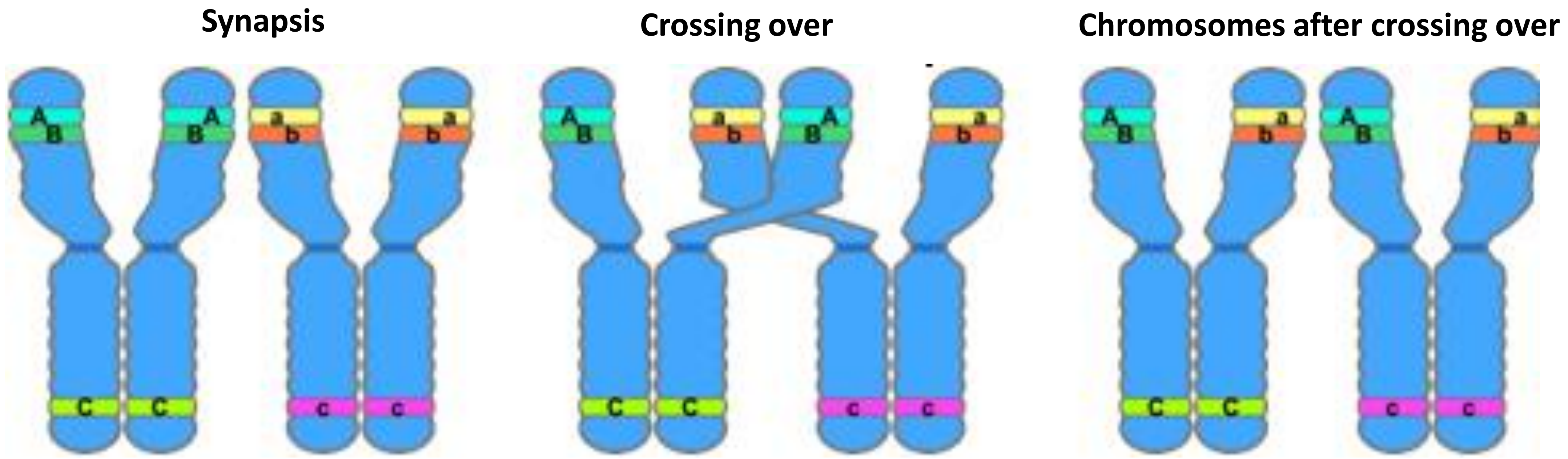
Sexual reproduction

- To reproduce, eukaryotes must first create special cells called gametes (eggs and sperms) that then fuse to form the beginning of a new organism.
- Meiosis is a specialized type of cell division that occurs during the formation of gametes. It is really just two cell divisions in sequence. Each of these sequences maintains strong similarities to mitosis.
  - Meiosis serves to reduce the chromosome number for that particular organism by half.


MEIOSIS
MITOSIS
DNA replication
Meiotic division 1
Homologous chromosomes at the same level on equatorial plate
CELL DIVISION 1
Meiotic division 2
CELL DIVISION 2
Homologous chromosome line up individually at the equatorial plate
CELL DIVISION

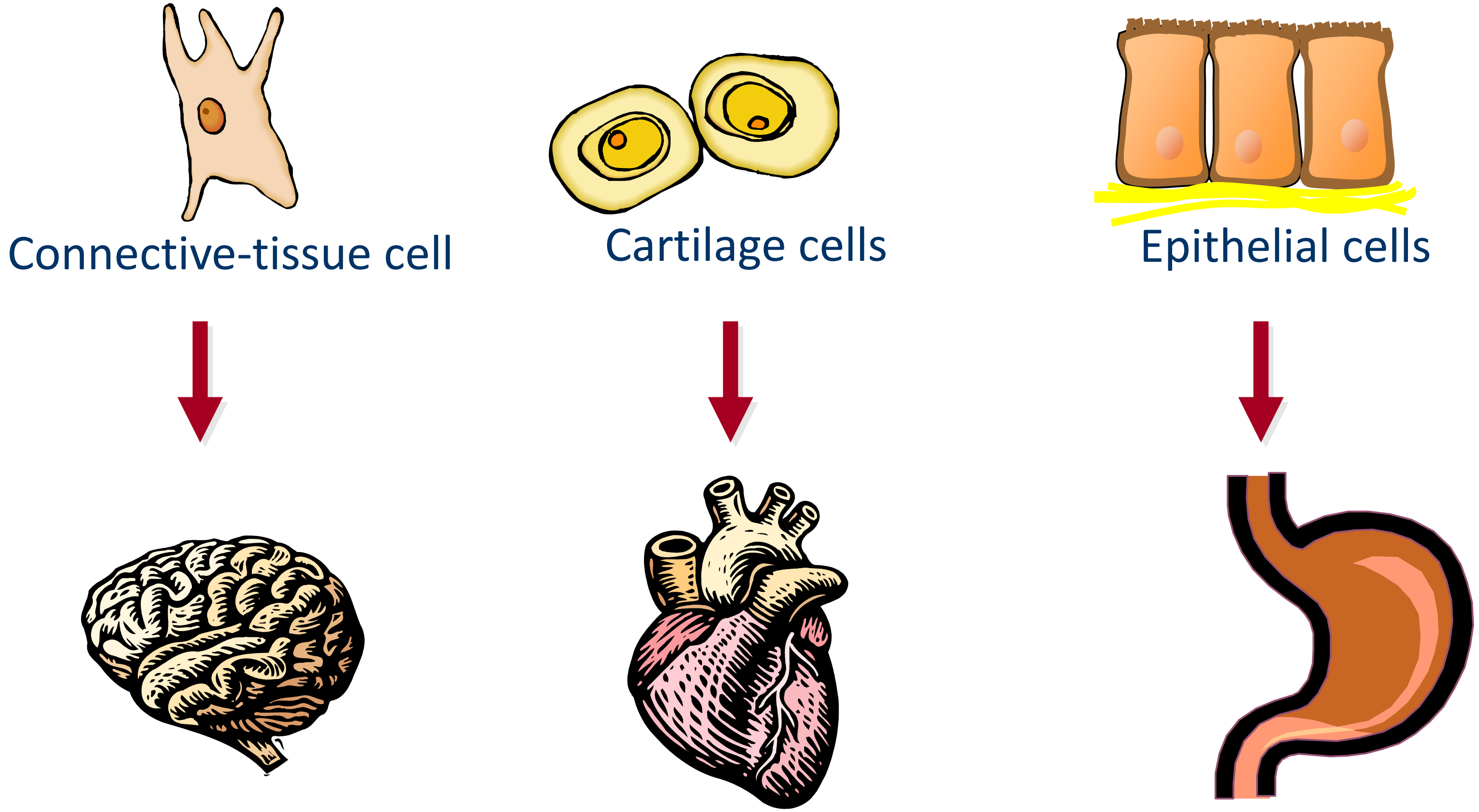
Genetic recombination - diversity

- Genetic recombination refers to a large-scale rearrangement of a DNA molecule.



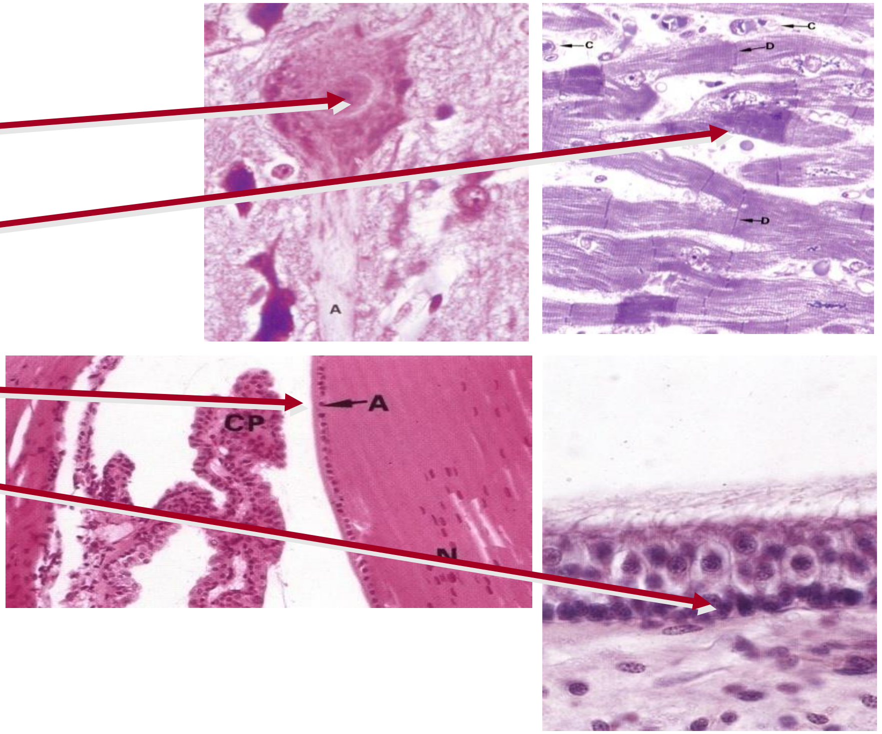
Cell differentiation

- **Differentiation** is the process by which an unspecialized cell becomes specialized into one of the many cells that make up the body



Cell differentiation (continued)

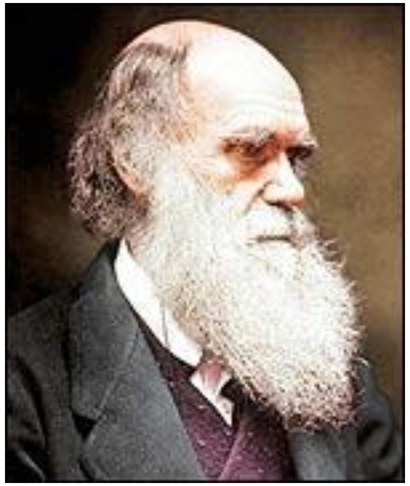
- During differentiation, certain genes are turned on, or become activated, while other genes are switched off, or inactivated
- This process is intricately regulated
- e.g.
  Nerve cells
  Cardiac muscle cells
  Lens cells of the eye
  Auditory hair cells of the ear



Appendix D: Genetic Variation

What is the difference between man and ape?

- Man and chimpanzee have a genomic similarity of greater than 95%
- What accounts for differences between species?



Physical traits and variances

- Individual variation among a species occurs in populations of all sexually reproducing organisms
- Individual variations range from hair and eye color to less subtle traits such as susceptibility to malaria
- Physical variation is the reason we can pick out our friends in a crowd. However, most physical traits and variation can only be seen at a cellular and molecular level

Genetic variation

- Despite the wide range of physical variation, genetic variation between individuals is quite small
- Out of 3 billion nucleotides, only roughly 3 million base pairs (0.1%) are different between individual genomes of humans
- Although there is a finite number of possible variations, the number is so high ($4^{3,000,000}$) that we can assume no two individual people have the same genome
- What is the cause of this genetic variation?

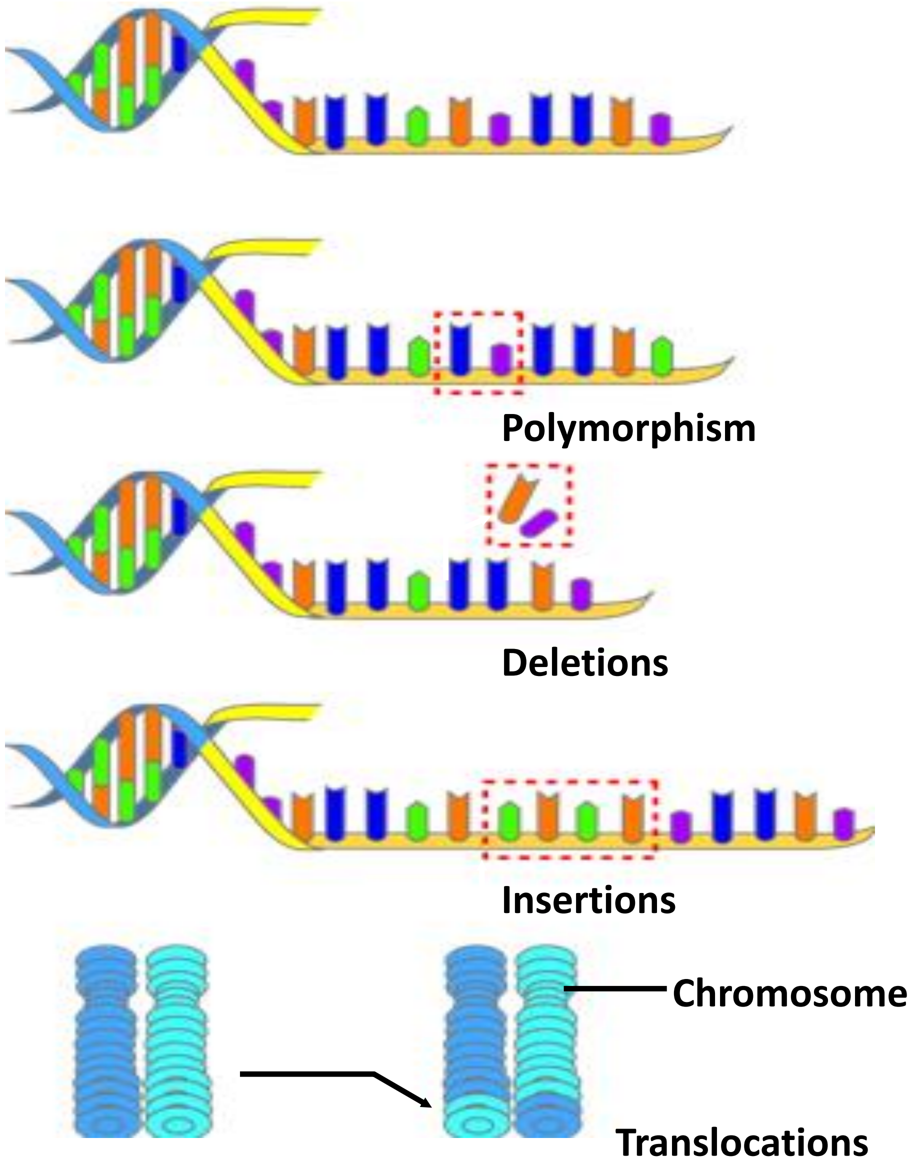
Sources of genetic variation

- **Mutations** are rare errors in the DNA replication process that occur at random
- When mutations occur, they affect the genetic sequence and create genetic variation between individuals
- Most mutations do not create beneficial changes and actually kill the individual
- Although mutations are the source of all new genes in a population, they are so **rare** that there must be another process at work to account for the large amount of diversity

Variation through mutation

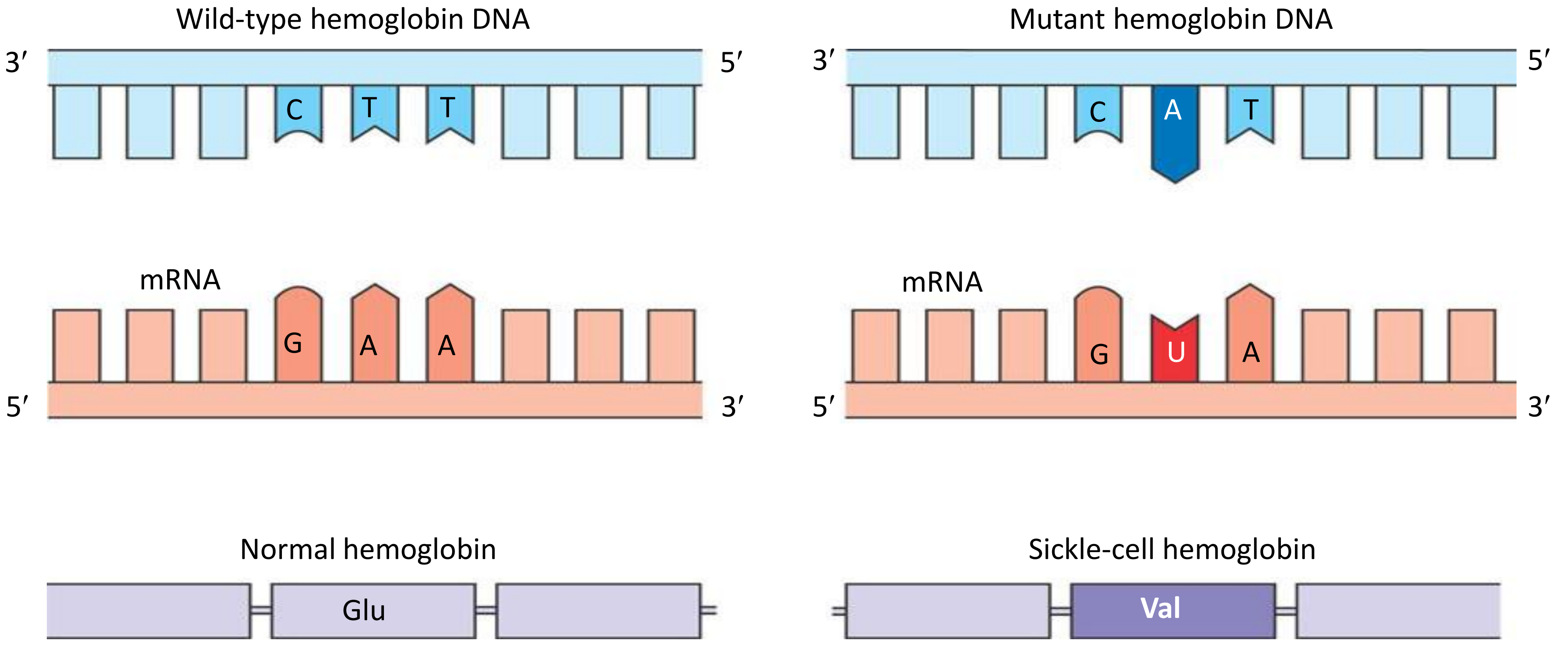
Common Sequence

Variations



- **Single Nucleotide Polymorphism** or **SNP**

The molecular basis of sickle-cell disease: a point mutation



In the DNA, the mutant template strand has an A where the wild-type template has a T.

The mutant mRNA has a U instead of an A in one codon.

The mutant (sickle-cell) hemoglobin has a valine (Val) instead of a glutamic acid (Glu).

More sources of genetic variation

- **Recombination** is the shuffling of genes that occurs through sexual mating and is the main source of genetic variation
- Recombination occurs via a process called **crossing over** in which genes switch positions with other genes during meiosis
- Recombination means that new generations inherit random combinations of genes from both parents
- The recombination of genes creates a seemingly endless supply of genetic variation within a species
- Other sources: genome rearrangement, gene duplication, etc.